

Structure and Response of Spherical Diffusion Flames

Principal Investigator: **Chung K. Law**
Princeton University

Co-Investigator: **Stephen D. Tse**
Rutgers University

Co-Investigator: **Kurt R. Sacksteder**
NASA-Glenn Research Center

SCR at NASA Glenn

February 20, 2008



Outline

- Goals
- Scientific significance and justification for μg experimentation
  - Merits of spherical diffusion flames
  - Elemental diffusion flame processes
  - Development of fundamental physical & chemical database
- Experimental plan

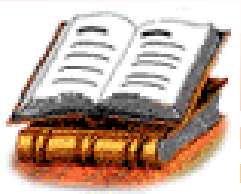


Goals

- Use spherical diffusion flames in long-duration μg environment to:
  - obtain experimental data of exceptional quality in simple, well-defined flow fields.
  - understand various key elemental diffusion flame processes.
- Compare these data with those obtained through independent computational simulation with detailed chemistry and transport in order to:
  - scrutinize and improve the accuracy and comprehensiveness of these descriptions.



Scientific Significance





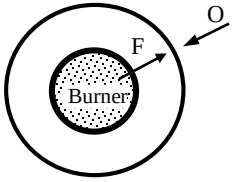
Merits of One-Dimensional Flames for Fundamental Studies

- Flame structure fundamentally affected by aerodynamics.
- Simple flow and flame configuration facilitate experimentation, computation, analysis, and comparison, with enhanced accuracy and reliability.
- Reduced effort in describing flows allows more detailed study of other aspects of flames such as chemistry.
- A one-dimensional flame is the simplest possible configuration.

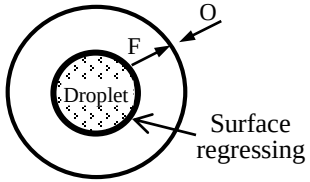


Merits of Burner-Generated Spherical Diffusion Flames

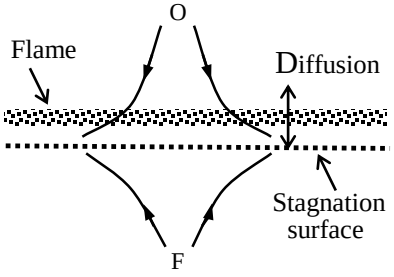
- Only spherical (vs. planar and cylindrical) configuration admits steady-state 1D behavior in semi-infinite domain.
- Spherically symmetric droplet combustion complicated by transient effects due to droplet heating and surface regression. Kinetics of large HC fuels also less well established.
- Burner generation assures steady-state surface boundary conditions.



Spherical Burner Flame



Droplet Flame



Counterflow Flame

- Planar flame in counterflow is not 1D in terms of diffusion-convection as well as radiation misalignment.



Program Components

FOCUS A

Experimentation on and understanding of elemental (diffusion) flame processes

FOCUS B

Development of fundamental chemical & physical database, especially comprehensive reaction mechanisms



Focus A: Experimentation on Elemental (Diffusion) Flame Processes

I. Spreading of diffusion flame sheets

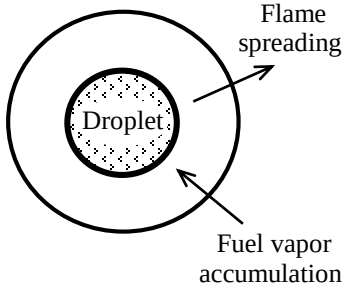
II. Dual extinction states

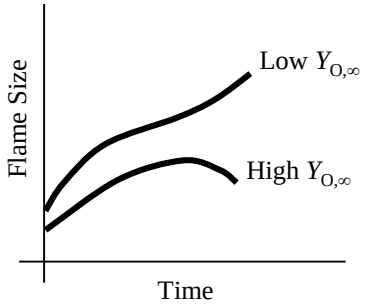
III. Flamefront instabilities

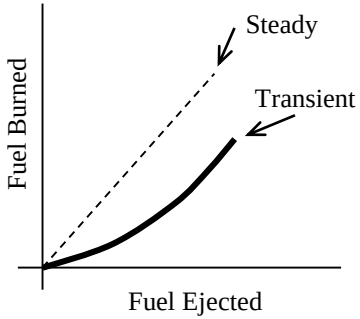


I. Spreading of Diffusion Flame Sheets: Phenomenology

- Steady-state assumption ignores initial conditions.
- An important process is fuel accumulation, i.e. scalar profile buildup, leading to spreading of the flame sheet from the fuel source.
- Relevant for all non-planar processes!







Droplet Flame Behavior

- Previous observation of droplet flame complicated by far field diffusion and surface regression.
- Insensitive to chemistry, allows focus on diffusion.
- Coupling to density and hence temperature fields allows assessment of radiation effects.



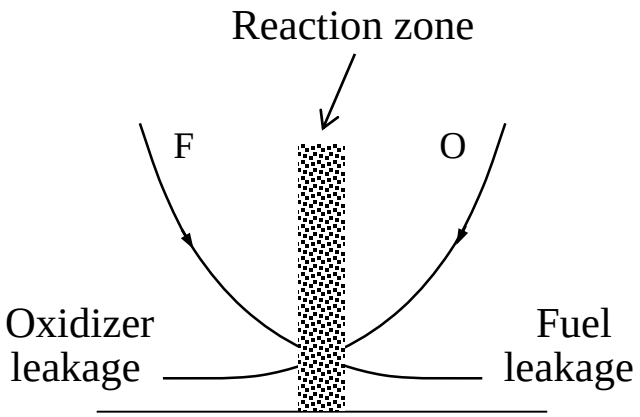
I. Spreading of Diffusion Flame Sheets: Experimental

- Fuel/diluent mixtures will be issued from the spherical burner and ignited at flow rates expected to form flames that evolve toward idealized steady-state conditions
- As the flame spreads outward, transient measurements of flame-front velocity and flame structure will be obtained using imaging and radiometric systems
- The flame characteristics will be compared with the predictions of the transient numerical simulation of spherical diffusion flames for evaluations of detailed chemical kinetics and radiation models.



II. Flame Extinction: Phenomenology (1/3)

- Unlike premixed flames, reactant leakage is the root cause of diffusion flame extinction.
- For thin reaction zones, effects of other loss processes (radiation, diffusional stratification, unsteadiness, multi-dimensional) can all be folded into reactant leakage through flame temperature reduction.



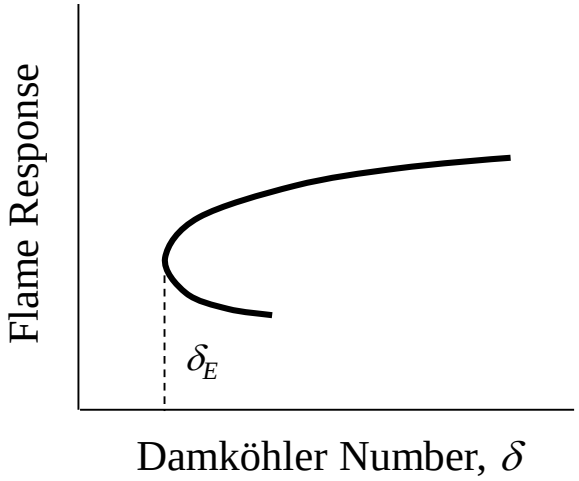


II. Flame Extinction: Phenomenology (2/3)

- Consequently practically all diffusion flame extinction phenomena have been found to be described by Liñán's canonical extinction criterion:

$$\delta < \delta_E$$

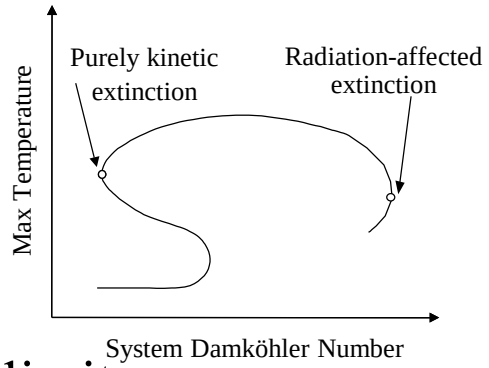
where δ is (Linán's) flame Damköhler number and $\delta_E \sim e(1-|\gamma|)$, $\gamma = \gamma$(freestream energy parameters).





II. Flame Extinction: Phenomenology (3/3)

- Flame can extinguish at both small & large Damköhler numbers
- Kinetically limited extinction occurs for small Da.
- Increasing Da implies longer residence time or large system dimension, leading to increased radiation loss, reduced flame temperature, and eventually extinction through excessive leakage



- Liṅán's extinction criterion applies at both limits
- Dual extinction observed in μg droplet burning (?)
- Present experiment further facilitates interpretation (*e.g.* no droplet heating, simpler fuels)



II. Flame Extinction: Experimental, Low Da

- Fuel/inert mixtures will be issued from the spherical burner and ignited at initial flow rates expected (or known/observed) to form flames that evolve toward idealized steady-state conditions
- Following ignition and initial flame front movement, the fuel/inert flow rate will be abruptly reduced to a second flow rate below the expected extinction limit
- Quasi-steady extinction is expected to occur as the flame adjusts to the reduced flow rate
- To refine the determination of the extinction flow rate for each fuel/diluent mixture, repeated tests are to be conducted at successively lower initial flow rates and higher secondary flow rates.



II. Flame Extinction: Experimental, High Da

- Fuel/inert mixtures will be issued from the spherical burner and ignited at flow rates slightly *above and beyond* that of the radiation extinction limit.
- Upon ignition the flame will grow and subsequently extinguish at the radiation extinction limit.
- Rates of radiative emissions from the flame and radiative reabsorption elsewhere are to be accentuated in some cases by using a radiatively active diluent such as CO_2.



III. Flamefront Instabilities: Phenomenology

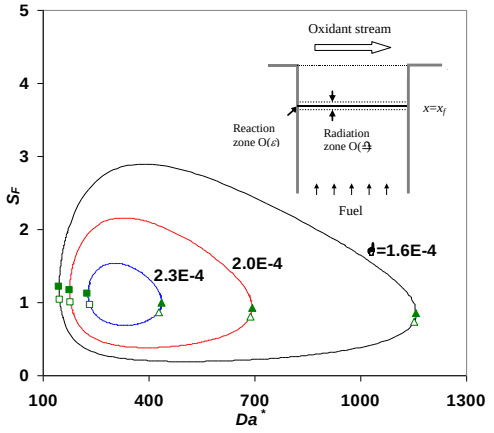
- Stoichiometry constraint and lack of chemistry render diffusion flame sheets absolutely stable.
- However, chemistry becomes important for near-limit burning. Will flame become unstable?
- Recent theories & experiments show instability characteristics similar to those of premixed flames for near-limit burning
- Since (positive) stretch suppresses cellular instability while promotes pulsating instability, the (stretchless) spherical flame is well suited for such studies.



III. Flamefront Instabilities: Experimental

- Manipulate fuel and oxidizer Le through inert substitution
  - Le<1: Cellularly unstable
  - Le>1: Pulsatingly unstable
- Radiating diluents are used to further promote instability

- Pulsating instabilities are predicted to occur near the extinction conditions at both low and high Da, thereby modifying the extinction mechanism and limit; also allows for extinction detection and flame control





Focus B: Development of Fundamental Physical & Chemical Database

- Detailed fuel oxidation mechanisms
- Reduced fuel oxidation mechanisms
- Diffusion coefficients
- Radiation models



Fundamental Needs in Computational Aerothermochemistry

- Satisfactory quantitative description of combustion phenomena depends on:
  - Accurate input of physics & chemistry
  - Computationally accessible
- Components of input:
  - Turbulence (DNS & modeled)
  - Chemistry (detailed & reduced)
  - Diffusion (mixture & second order)
  - Radiation (various models)

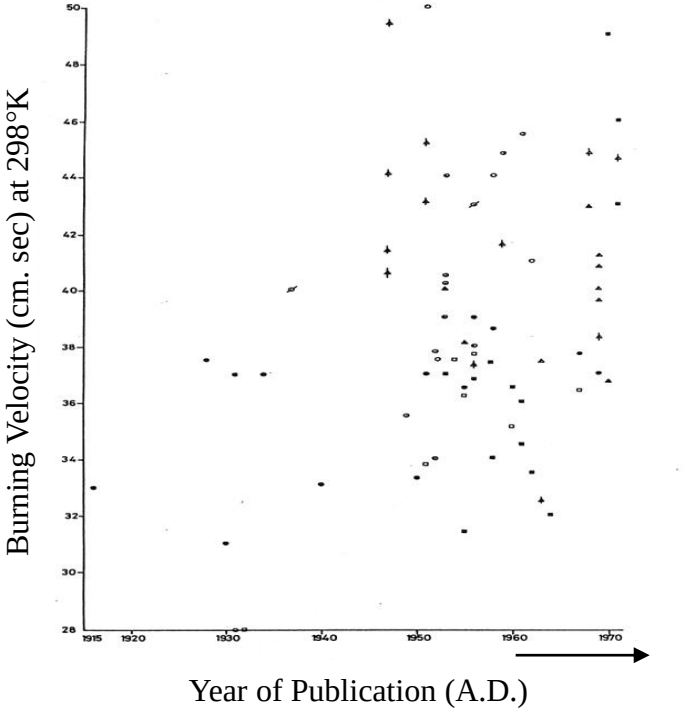


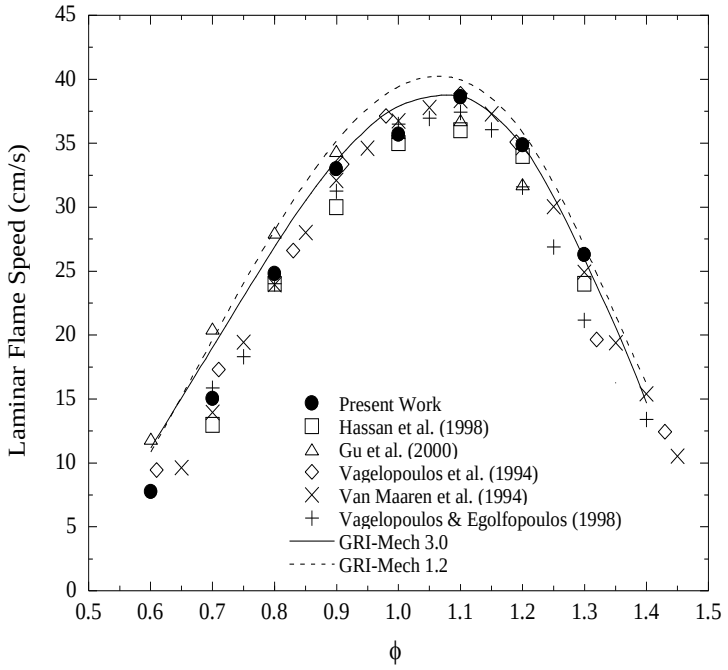
Complexities in Description of Chemistry

- Fuel oxidation is highly nonlinear and nonmonotonic in terms of temperature, pressure, and species concentrations
- Oxidation mechanisms consist of large number of species and reactions
- Characteristic reaction rates span over many orders of magnitude, hence computationally stiff
- Multicomponent diffusive transport can also be computationally expensive



Importance of Data Fidelity: Methane-Air Flame Speeds





ϕ=1 data by 1970s;

Stretch contaminated

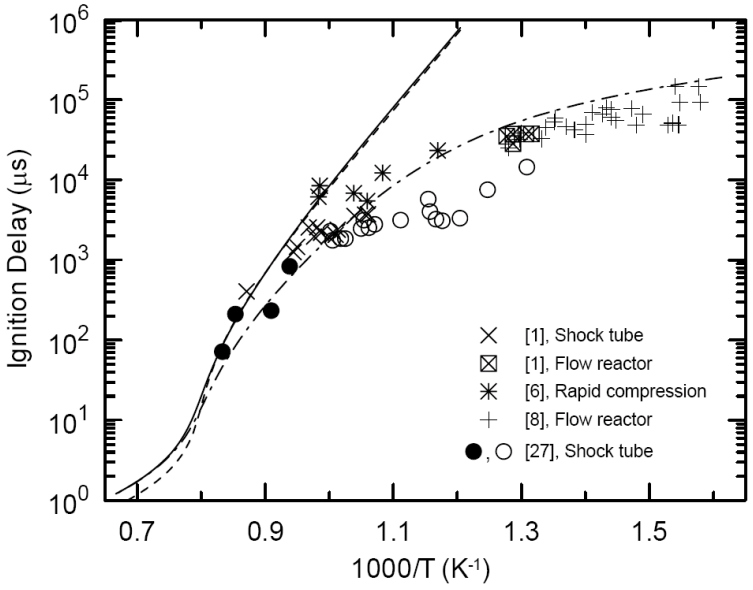
1 atm data after 1980s;

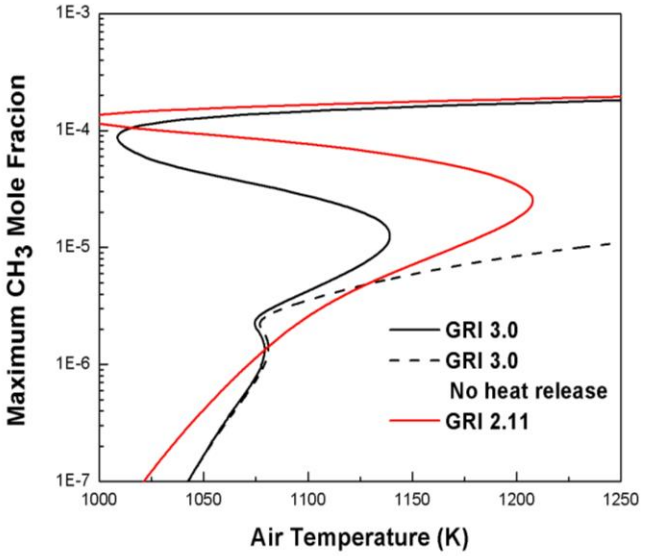
Stretch effect eliminated



(Inadequate) State of Detailed Mechanisms

- Hydrogen: Disagreement between calculation & experiment in ignition delay at low temperatures & high pressures
- Methane: Disagreement in the nature of ignition for calculations with different mechanisms (GRI-Mech 2.11 vs 3.0)







Targets for Comprehensiveness in Experimental Data

- Broad coverage of thermodynamic states: temperature, pressure, concentration
- Ignition (homogeneous and diffusive): low-temperature chemistry
- Steady burning & extinction: high-temperature chemistry
- Premixed & diffusion flames: mixedness & time scale
- Global responses (flame speeds and extinction strain rates)
- Detailed flame structure (thermal and major/minor species)
- Pollutant chemistry



The Role of μg Spherical Diffusion Flame in Comprehensive Mechanism Development

- Substantial research on homogeneous systems (shock tubes, flow reactors, *etc*.) and premixed flames
- Diffusion flame experiments mostly involve counter- and stagnation flows, and droplet flames
- Burner-supported spherical flame is 1D and steady state
- Experiments of interest to chemistry are those involving ignition and extinction.
- Experiments for situations away from extinction would yield information on the diffusion and radiation parameters.



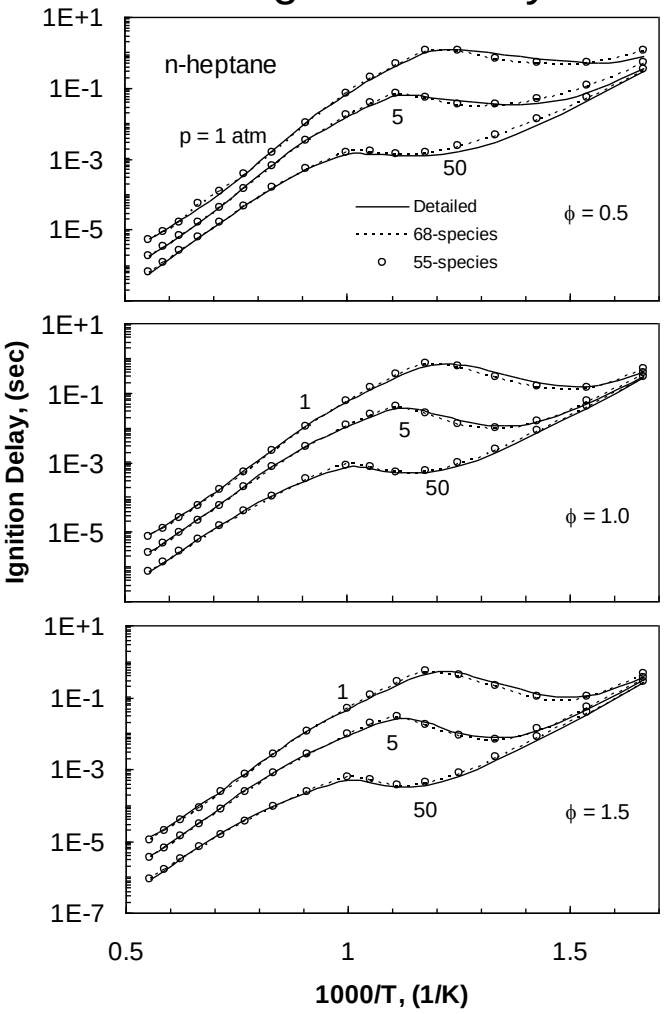
Development of Reduced Mechanisms

- Methodologies mostly in place for fast, systematic & accurate mechanism reduction
  - Elimination of unimportant species & reactions
  - Time-scale analysis (steady-state and partial equilibrium assumptions)
  - Diffusion coefficient lumping
- Example: Detailed n-heptane mechanism with 561 species and 2539 elementary reactions reduced to 55 species and 51 lumped reactions

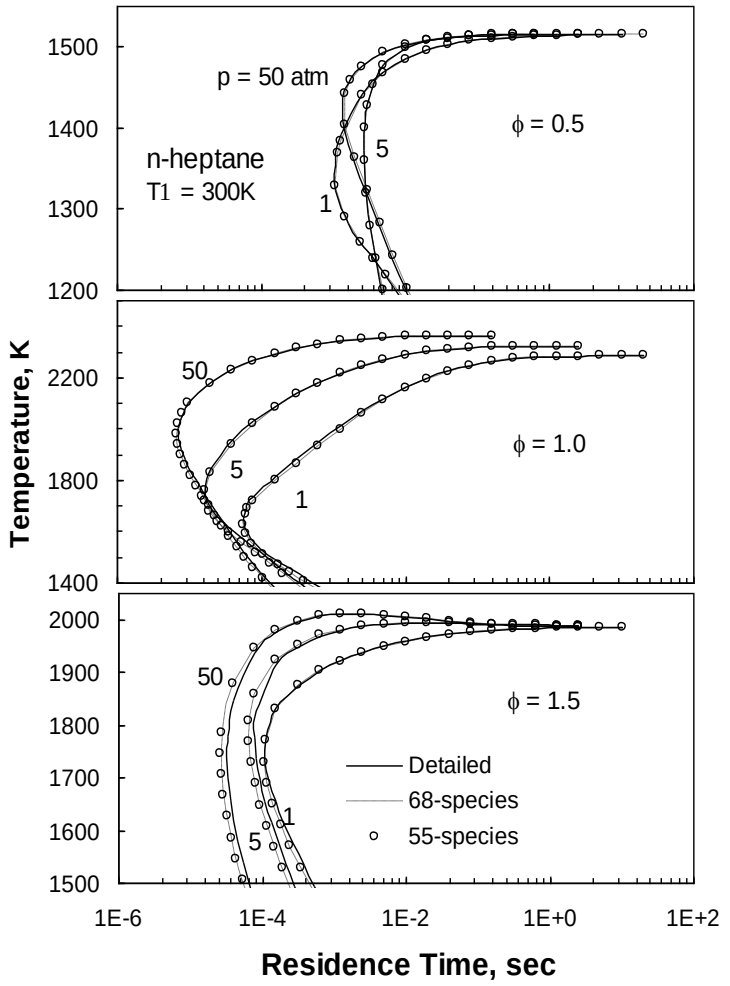


Validation of Reduced Heptane Mechanism

Ignition Delay



PSR Extinction





Summarizing Comments

- μg experimentation facilitates understanding of three important elemental flame processes:
  - Flame spread and fuel accumulation
  - Dual extinction limits: kinetic vs. radiation affected
  - Flame-front instabilities and extinction monitoring
- Acquisition of diffusion flame data from the clean and well-controlled μg experiments allow scrutiny and development of physical and chemical database, especially comprehensive detailed reaction mechanisms.



Structure and Response of Spherical Diffusion Flames

Principal Investigator: **Chung K. Law**
Princeton University

Co-Investigator: **Stephen D. Tse**
Rutgers University

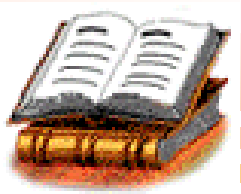
Co-Investigator: **Kurt R. Sacksteder**
NASA-Glenn Research Center

SCR at NASA Glenn

February 20, 2008

s-Flame

Experimental Objectives





Experimental Objectives

- A series of experiments is proposed, in both soot-free and sooty flames, to compare with detailed computational simulation to assess transport properties and detailed chemical kinetics (including chemiluminescence).



Experimental Objectives

Objective A. Measure and characterize the transient structure (such as temperature and species distribution) of diffusion flames established at initially non-steady, spherical flame structures for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty), for different flow rates.

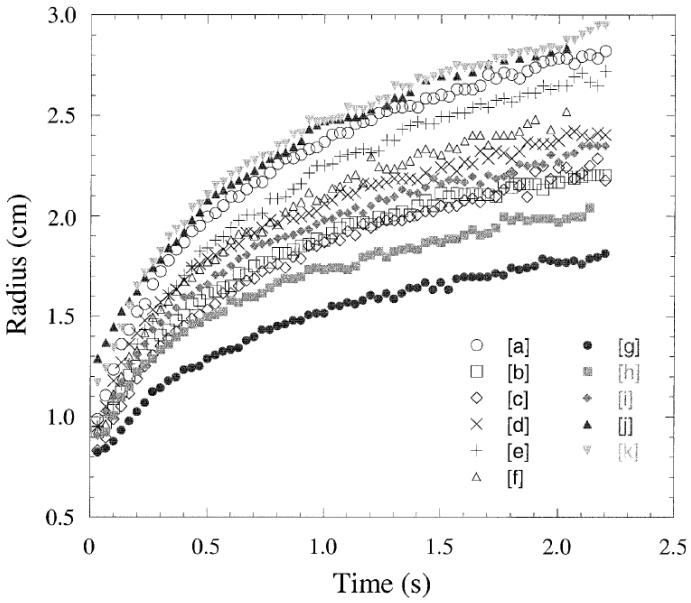
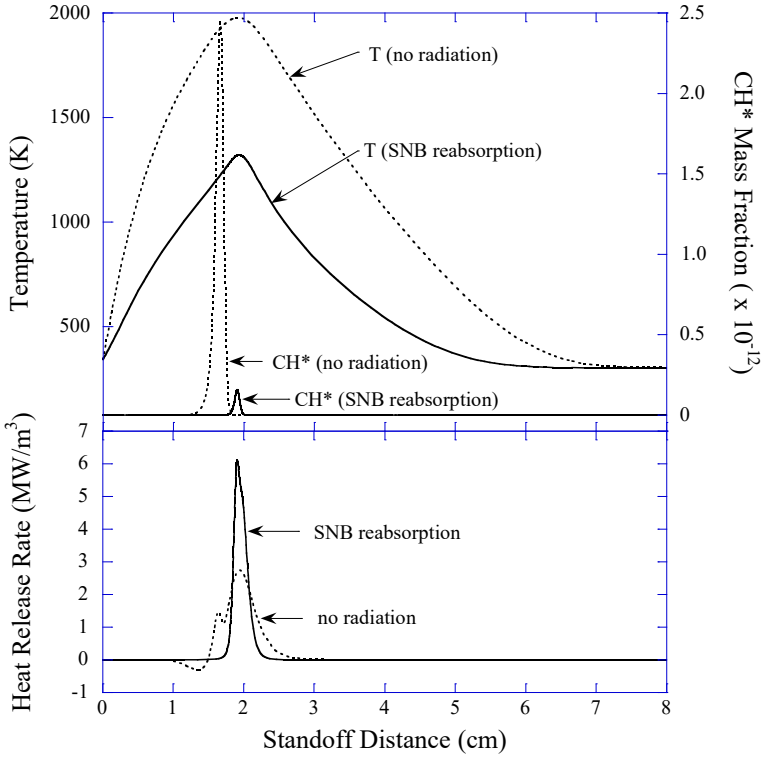
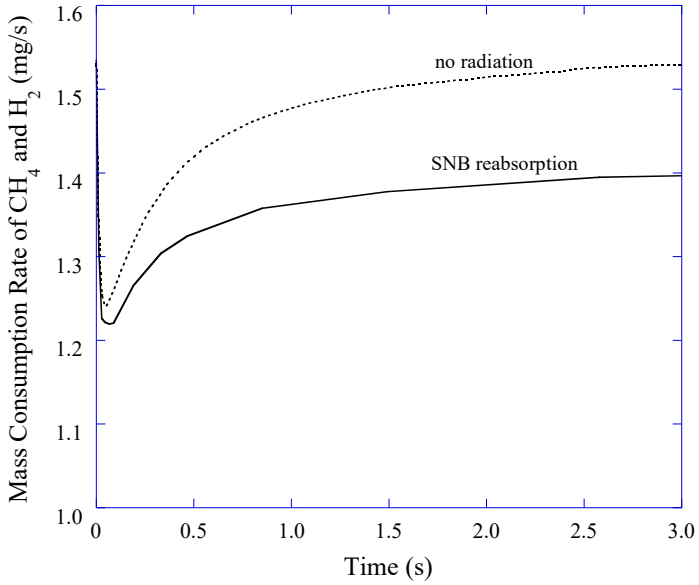


Fig. 3. Data of flame radii as a function of time for various cases. *20%H_2-25%CH_4-55%N_2*: [a] 15.72 mg/s. *25%H_2-20%CH_4-55%N_2*: [b] 10.55 mg/s. *30%H_2-15%CH_4-55%N_2*: [c] 9.42 mg/s, [d] 12.0 mg/s, [e] 15.31 mg/s. *40%H_2-10%CH_4-50%N_2*: [f] 12.37 mg/s. *50%H_2-10%CH_4-40%N_2*: [g] 4.96 mg/s, [h] 6.43 mg/s, [i] 8.1 mg/s, [j] 1.67 mg/s, [k] 13.03 mg/s.



Flame Structure Spread

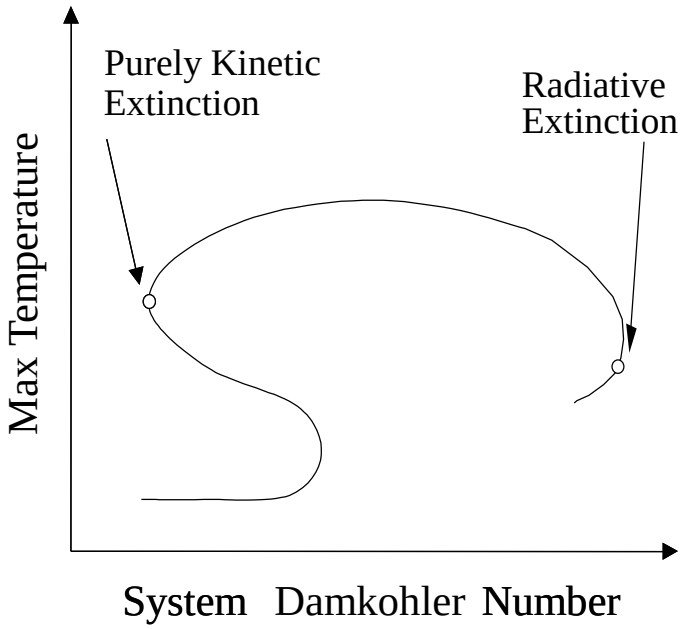






Experimental Objectives

Objective B. Determine the quasi-steady convective/chemical-kinetic extinction limits (low system Damkohler numbers) and radiative/chemical-kinetic extinction limits (high system Damkohler numbers) for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty).





Quasi-Steady Equations

The quasi-steady conservation equations, in the reference frame moving with the reaction front, for mass, energy, and species are given by:

$$\frac{\partial \dot{m}}{\partial x} + 2\frac{A}{r}\rho U_f = 0,$$

$$\dot{m}\frac{\partial T}{\partial x} - \frac{1}{c_p}\frac{\partial}{\partial x}\left(A\lambda\frac{\partial T}{\partial x}\right) + \frac{A}{c_p}\sum_{k=1}^{K}\rho Y_k V_k c_{p_k}\frac{\partial T}{\partial x} + \frac{A}{c_p}\sum_{k=1}^{K}\dot{\omega}_k h_k W_k = 0$$

$$\dot{m}\frac{\partial Y_k}{\partial x} + \frac{\partial}{\partial x}\left(\rho A Y_k V_k\right) - A\dot{\omega}_k W_k = 0$$

where

$$\dot{m} \equiv \rho(u - U_f)A \qquad A \equiv 4\pi r^2 \qquad \mathrm{d}x \equiv \mathrm{d}r - U_f\,\mathrm{d}t$$

U_f is the moving velocity of the flamefront.

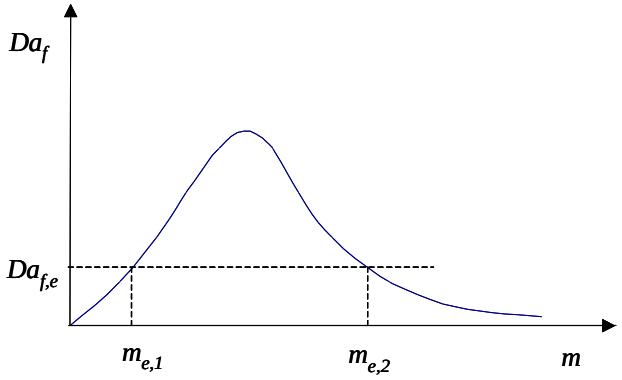


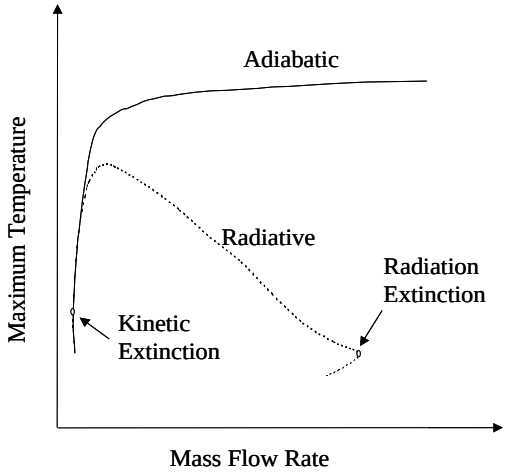
Analysis of Dual Extinction Turning Points

- $Da_f \sim m^2 \exp(-Am^2)$
- For given extinction Da_f, $Da_{f,e}$, there are two roots:

  $m_{e,1}$: Pure kinetic limit

  $m_{e,2}$: Radiation-affected limit



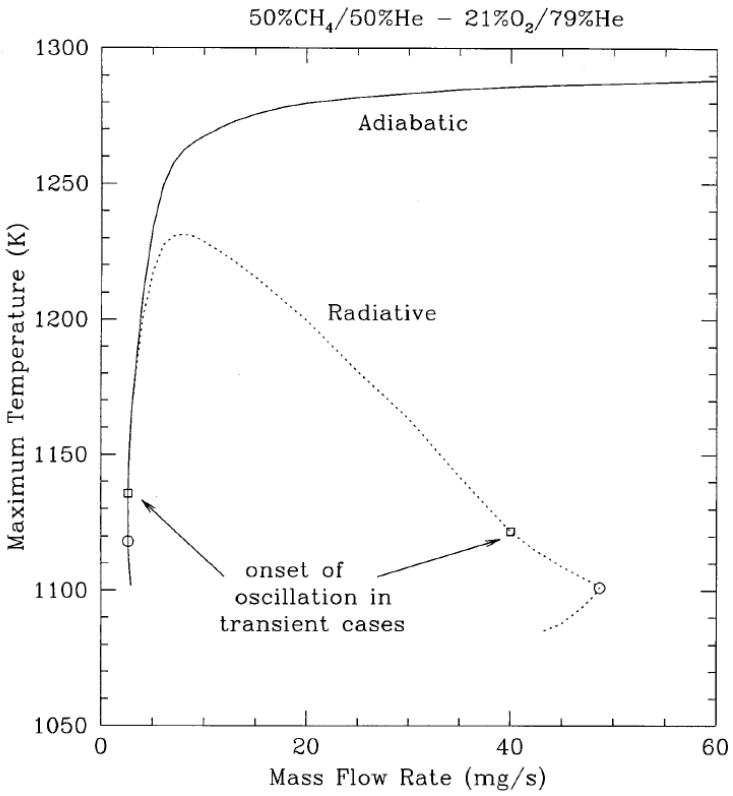


Maximum temperature in steady-state spherical methane diffusion flames versus mass flow rate, both adiabatic and with radiative heat loss.



Experimental Objectives

Objective C. Determine the existence (for different Zeldovich number, Ze), onset (as a function Da), and nature (i.e. mode, frequency, amplitude) of pulsating instabilities theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Lewis number.

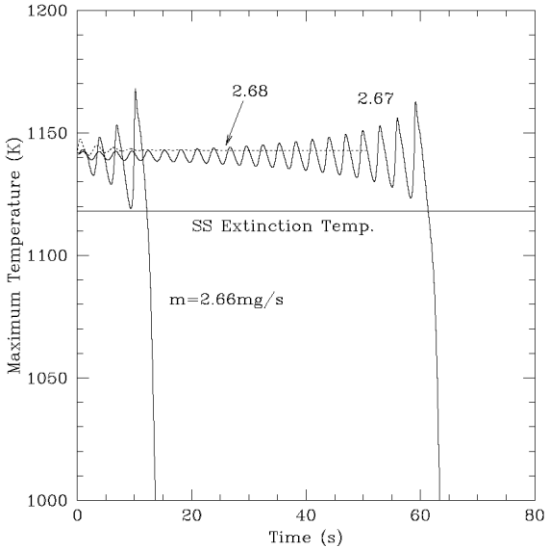




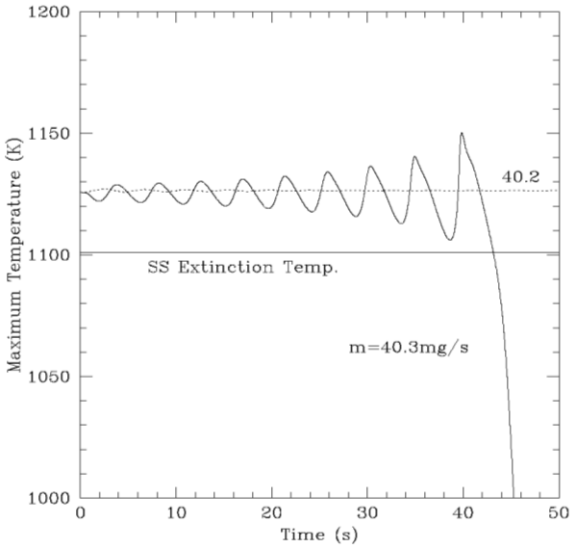
Oscillatory Response Prior to Extinction

- 50%CH_4 / 50%He vs. air (Le_F = 2.3, Le_o = 1.9)
- Oscillation observed at both kinetic and radiation extinction limits

Kinetic Extinction



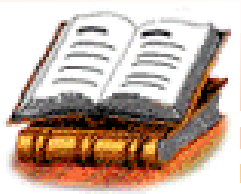
Radiation Extinction



Computation



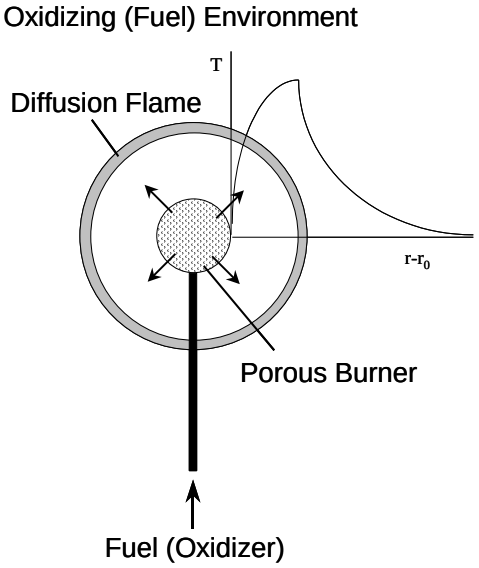
Proposed Experimental Approach

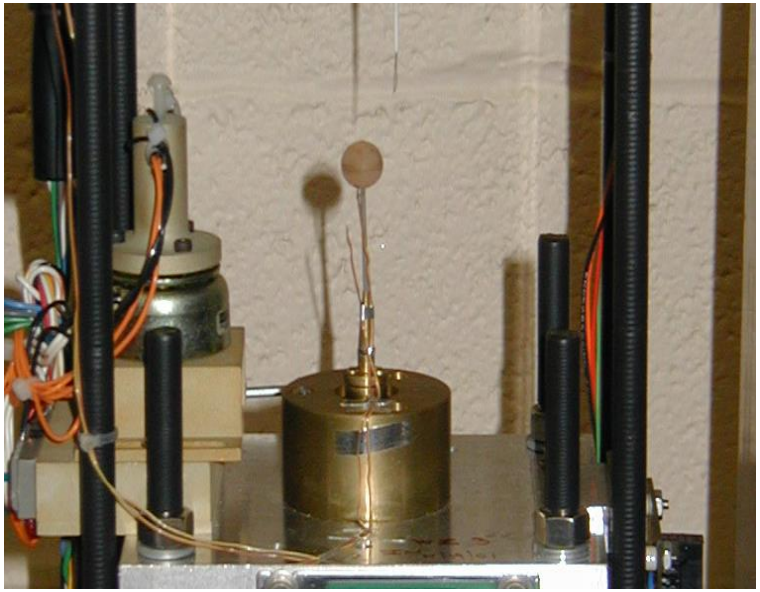




Experimental Approach (1/6)

- A series of experiments is proposed in which mixtures of various gaseous fuels, diluents and oxygen will be issued into an oxidizing environment through a spherical porous burner and ignited to nominally obtain a spherical, one-dimensional stabilized flame.







Experimental Approach (2/6)

- **Objective A** – Flame Structure Evolution
  - Fuel/diluent mixtures will be issued from the spherical burner and ignited at flow rates expected to form flames that evolve toward idealized steady-state conditions.
  - As the flame spreads outward, transient measurements of flame front velocity and flame structure will be obtained using imaging and radiometric systems.
  - The flame characteristics will be compared with the predictions of the transient numerical simulation of spherical diffusion flames for evaluations of detailed chemical kinetic and transport models.



Experimental Approach (3/6)

- **Objective B** – Quasi-Steady Extinction Limits, Low Da
  - Fuel/inert mixtures will be issued from the spherical burner and ignited at initial flow rates expected (or known/observed) to form flames that evolve toward idealized steady-state conditions.
  - Following ignition and initial flame front movement, the fuel/inert flow rate will be abruptly reduced to a second flow rate below the expected extinction limit.
  - Quasi-steady extinction is expected to occur as the flame adjusts to the reduced flow rate.
  - To refine the determination of the extinction flow rate for each fuel/diluent mixture, repeated tests are to be conducted at successively lower initial flow rates and higher secondary flow rates.



Experimental Approach (4/6)

- **Objective B** – Quasi-Steady Extinction Limits, Low Da
  - At the low Da extinction limit, chemical kinetic rates are retarded with respect to flow rates by inert gas in the fuel stream that decreases the flame temperature.
  - Based on theoretical predictions, fuel/inert mixtures are to be chosen for which the extinction flame radius is large enough to minimize the effects of heat loss to the burner.
  - The surface temperature of the burner will be directly measured to quantify this loss.



Experimental Approach (5/6)

- **Objective B** – Quasi-Steady Extinction Limits, High Da
  - Fuel/inert mixtures will be issued from the spherical burner and ignited at flow rates *above and beyond* those expected (or known/observed) to form steady flames when radiative loss is considered.
  - The quasi-steady extinction is expected to occur as the flame moves outward before reaching the *adiabatic* steady-state radius.
  - Rates of radiative emissions from the flame and radiative reabsorption elsewhere are to be accentuated in some cases by using a radiatively active diluent such as CO_2.

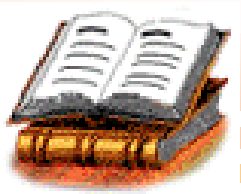


Experimental Approach (6/6)

- **Objective C** – Flame Front Instabilities
  - Pulsating flame front instabilities are predicted to occur near the extinction conditions at both low and high Da.
  - The approach taken to observe the instabilities is similar to the approach for observing extinction except fuel/diluent mixtures with large system Lewis numbers (Le) are used.
    - Low Da: moderate initial flow rate, abrupt change to flow rate below predicted extinction rate.
    - High Da: initial flow rate above predicted extinction flow rate with radiation.
    - Instabilities are expected to appear before extinction.
  - Radiating diluents are used to further encourage the instabilities.

s-Flame

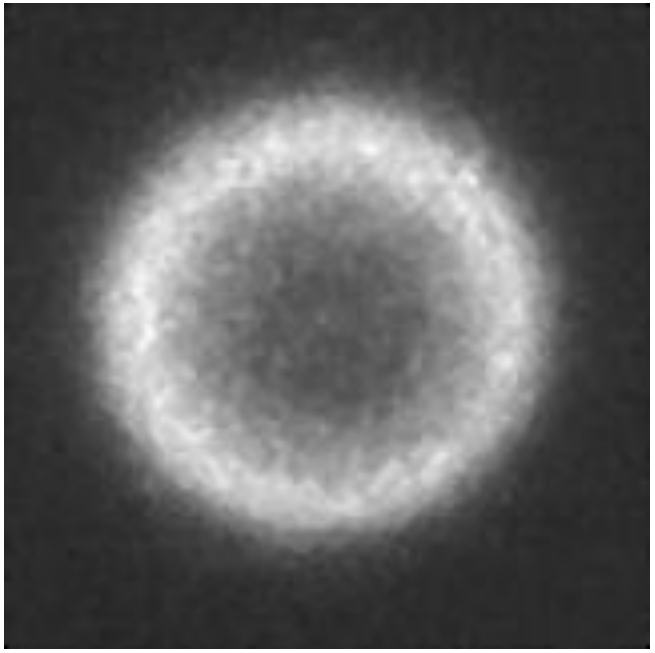
Justification for Extended Duration Microgravity



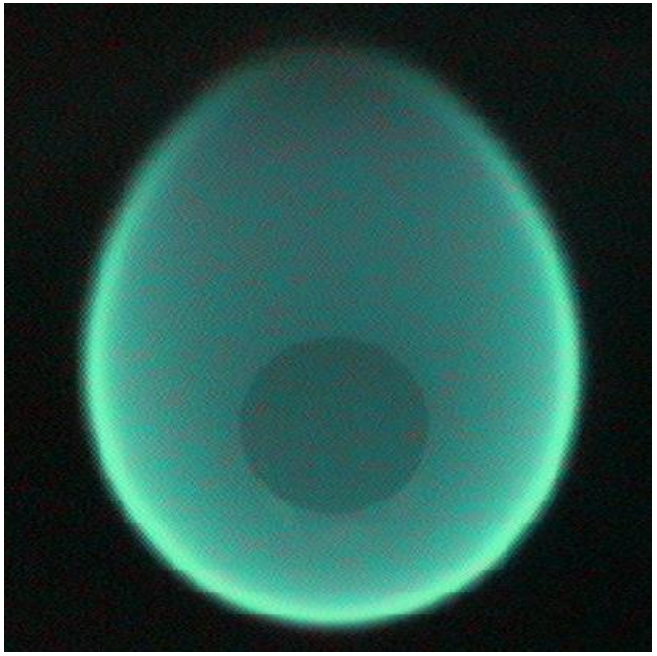


Limitations of Testing in n-*g*

- Need truly one-dimensional flames
- Inverse-flame configuration is limited and constrained



Hydrogen

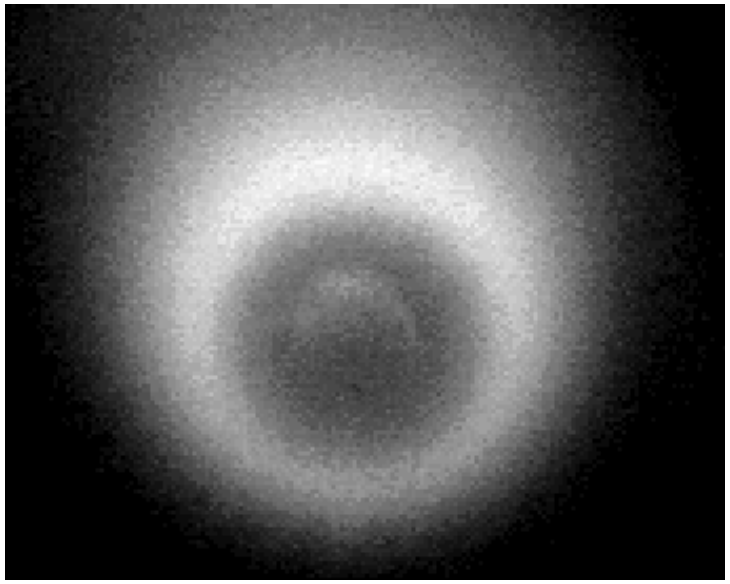


Methane



Limitations of Testing in n-*g*

- Thermal radiation from CO_2 (4.31μm) and H_2O (1.87μm)
  - Flame structure and equilibrium chemistry



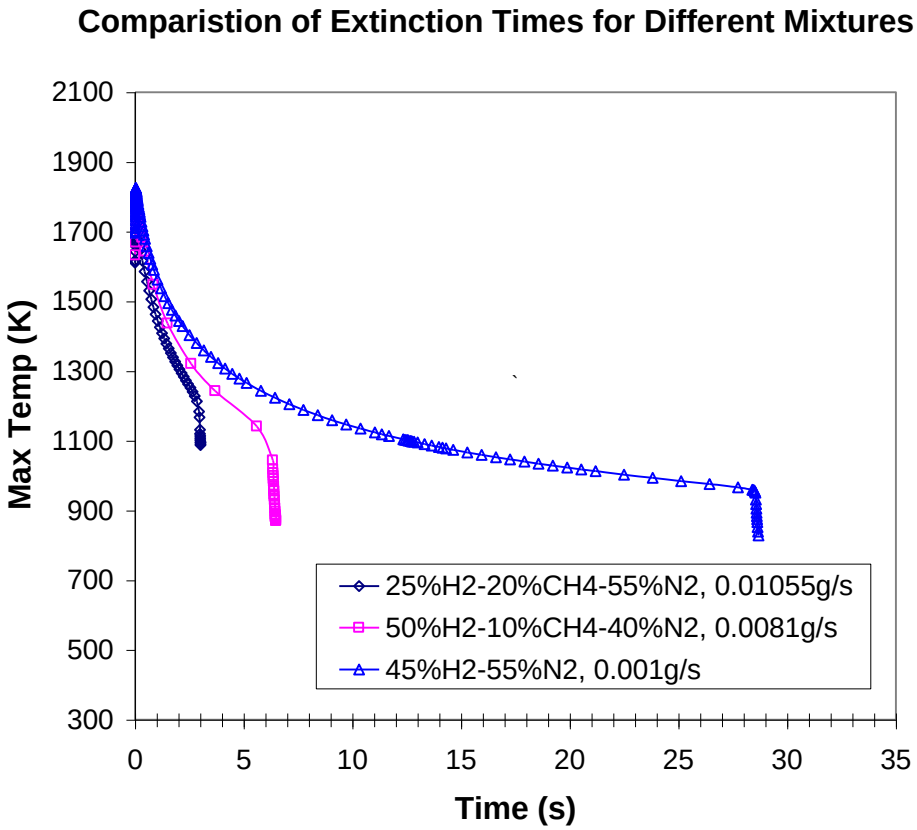
Projection image (2.81um)

- 9%O_2/91%N2, m = 0.078 g/s, into 100%H_2 at 0.079 atm
- AGEMA Thermovision 900 infrared camera with filter
- H_2O emission



Limitations of Drop Towers

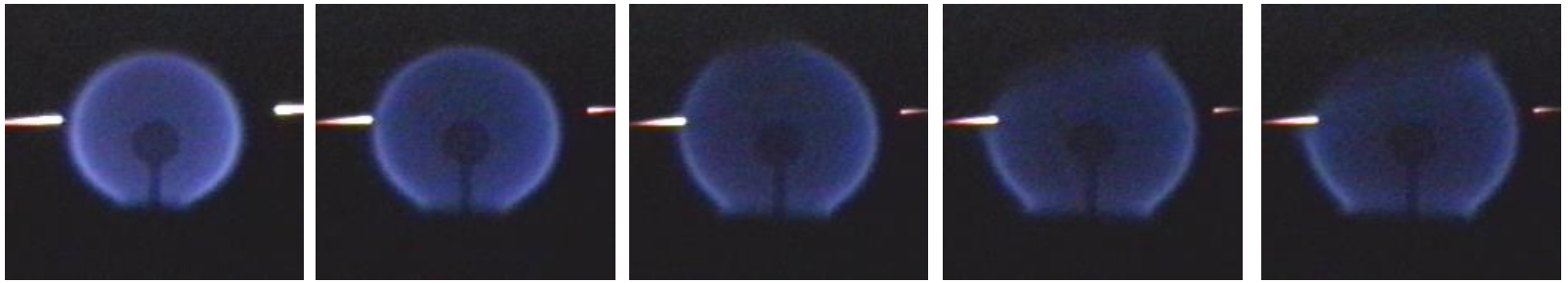
- Durations in 2.2-s and 5.0-s drop towers insufficient to observe radiative extinction





Limitations of Drop Towers

- Preliminary indications of local radiative extinction observed in 5s drop tower, but
  - Need more time to observe quasi-steady flame spread prior to TOTAL radiative extinction
  - Multi-staged flames → kinetic extinction

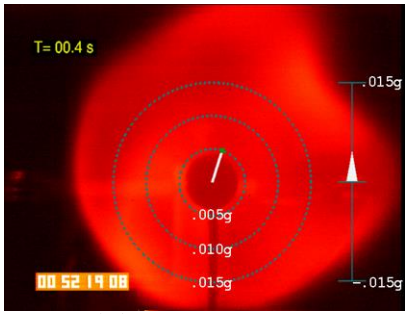


20%H_2 / 25%CH_4 / 55%N_2

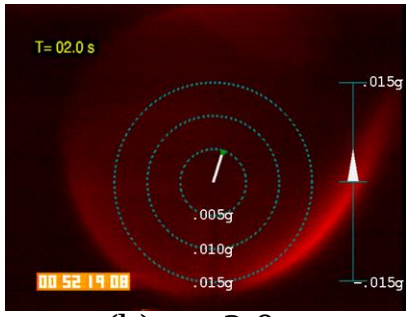


Limitations of Aircraft

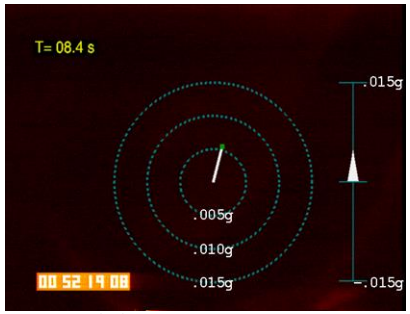
- Radiative extinction never observed in parabolic aircraft (up to 10s low-*g*) → *g*-jitter, $Gr \sim l^3$



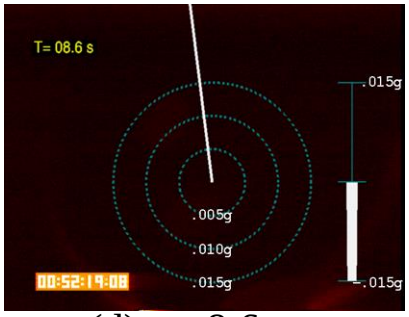
(a) t = 0.4 sec



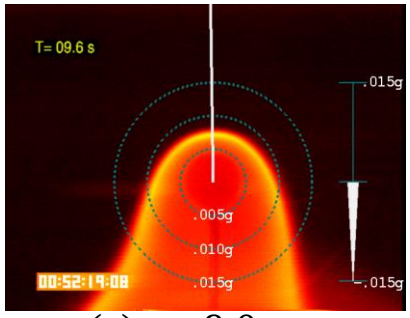
(b) t = 2.0 sec



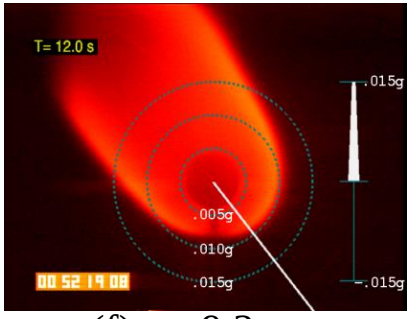
(c) t = 8.4 sec



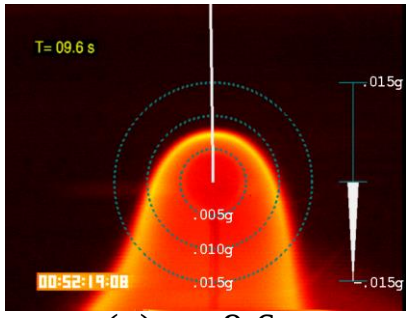
(d) t = 8.6 sec



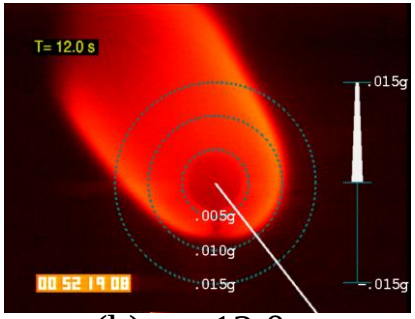
(e) t = 9.0 sec



(f) t = 9.3 sec



(g) t = 9.6 sec



(h) t = 12.0 sec

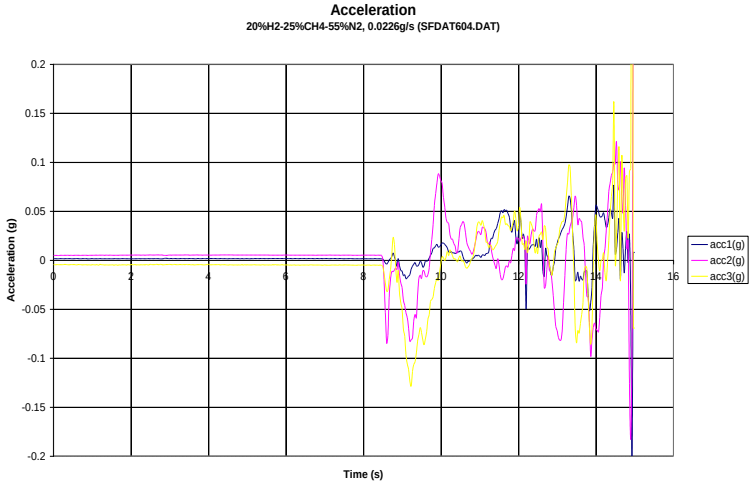
Fuel: 20%H_2/25% CH_4/ 55% N_2 @ 22.6 mg/sec in Ambient Dry Air

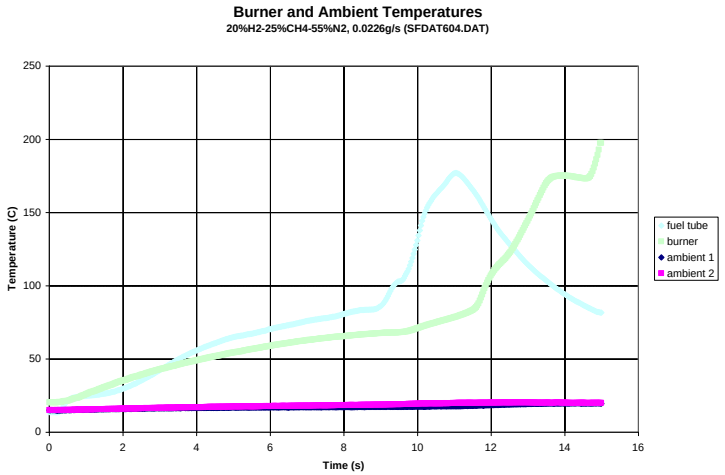


Limitations of Aircraft

Aircraft Experimentation

- Over 120 free-floated hardware experiments
- No extinction observed

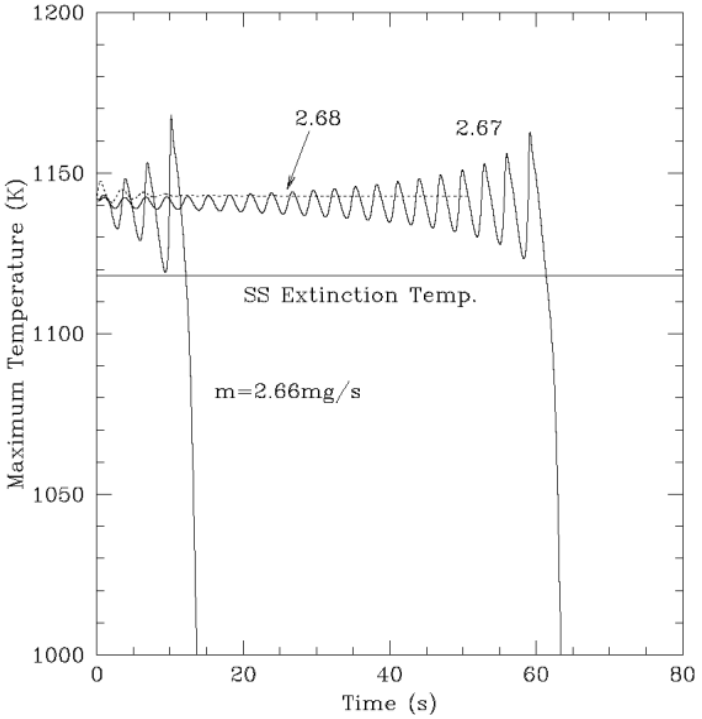






Limitations of Drop Towers and Aircraft

- Need extended duration and high-quality microgravity to study diffusion flame oscillations with “low” characteristic frequency



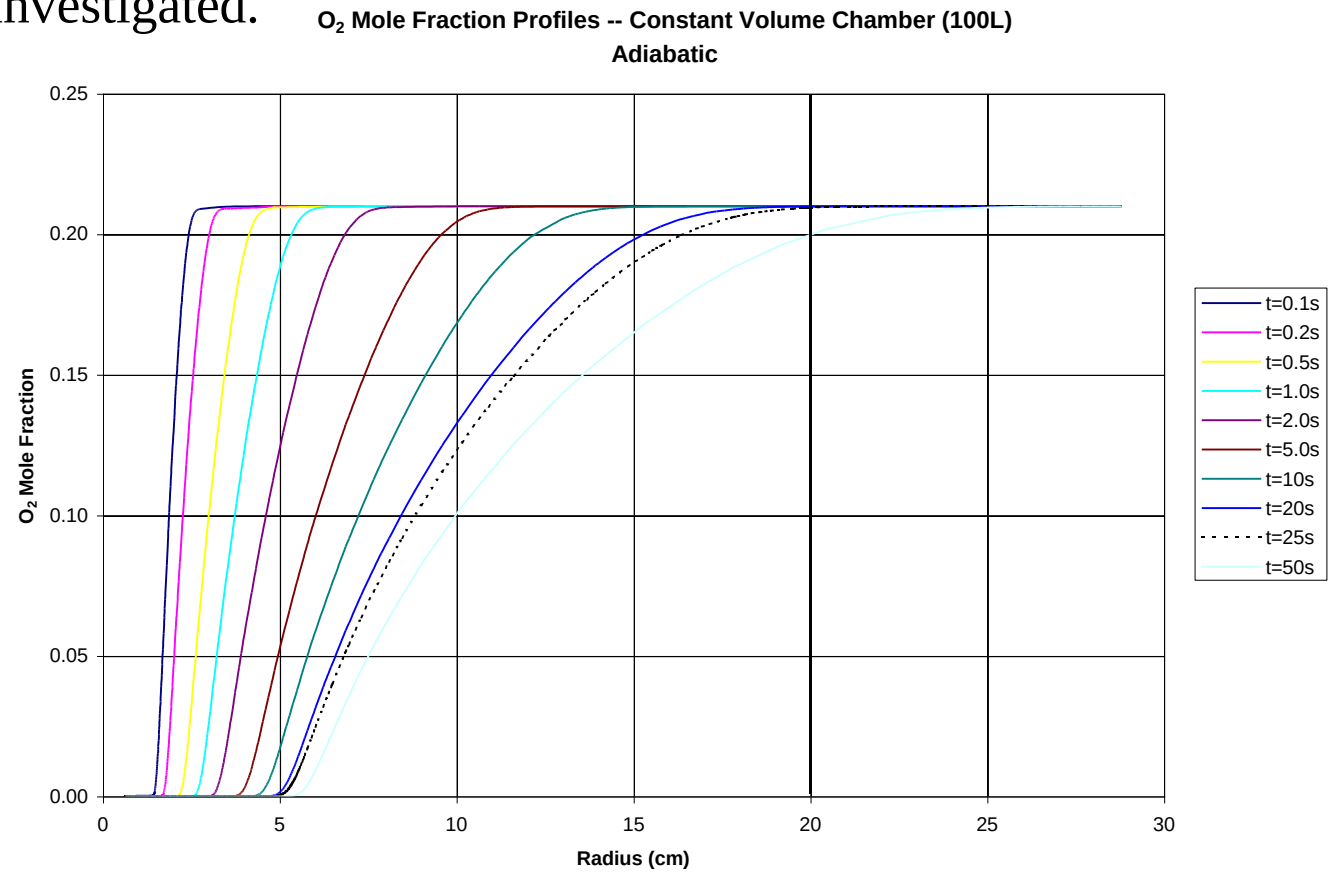
Pulsating Instability at the Low Mass Flow Rate Limit

50%CH_4 / 50%He fuel stream into 21%O_2 / 79%He ambient



Justification for Space-Flight Facility

- Estimated that 25s is sufficient to observe the processes investigated.



20%H_2-25%CH_4-55%N_2 into air, m=15.72mg/s

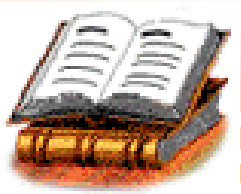


Justification for Space-Flight Facility

- High-quality and sufficient duration microgravity
  - Transient evolution to “steady-state”
  - Radiative extinction
  - Flame instability
  - Multi-staged flow for kinetic extinction
- Computation comparison
- Diagnostics afforded, KEY → Flame structure
  - UV imaging
  - Radiometer
  - Temperature profile

s-Flame

Science Data End Products





SDEP

Objective A) **Flame Structure Evolution**: Measure and characterize the transient structure of diffusion flames established at initially non-steady, spherical flame structures for mixtures of different H2/CH4/diluent ratios (soot-free) and different C2H4/diluent ratios (sooty), for different flow rates.



SDEP A1

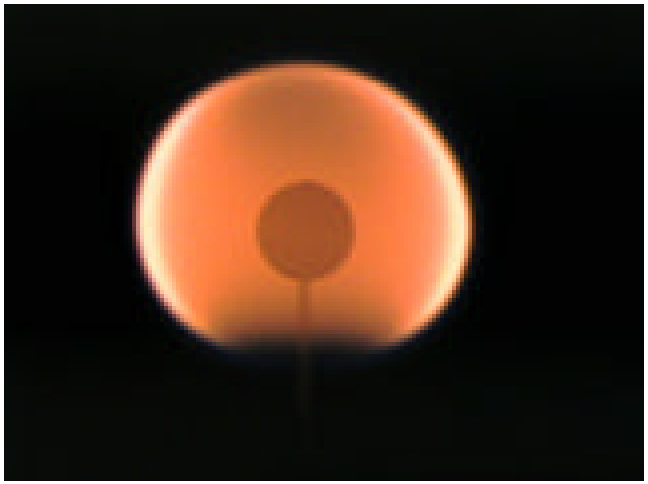
- Sequence of color and OH* images of the flame as a function of time, flow rate, and fuel composition.
- Requirements
  - Color images
  - UV images



Direct Color (Visible) Imaging

- Visible Chemiluminescence (CH*)
- Soot Incandescence
- Assess flame symmetry and influence of feed tube

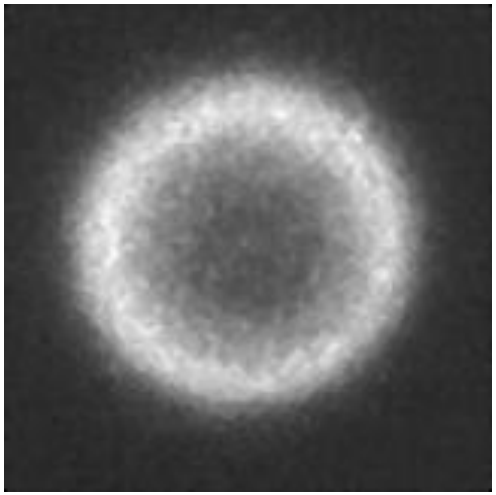




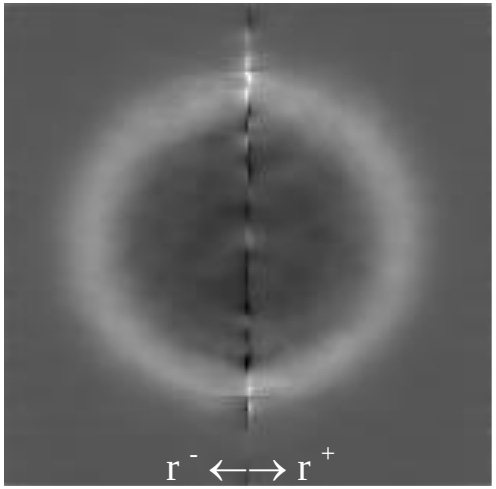


UV Imaging

- Chemiluminescence from OH* (310nm) or CH* (431nm)
  - Chemistry
  - Indirect assessment of flame structure



Projection Image (310nm)



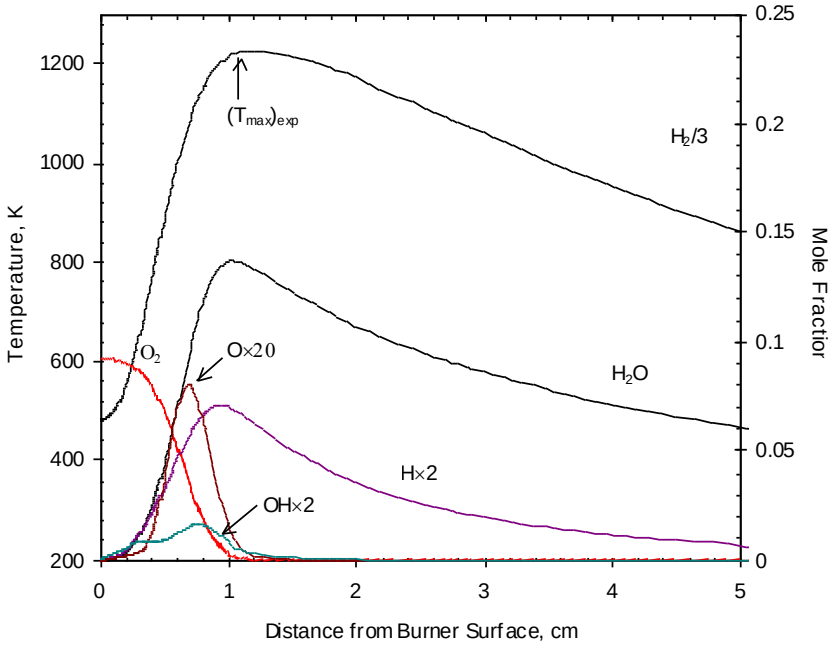
Abel deconvoluted image

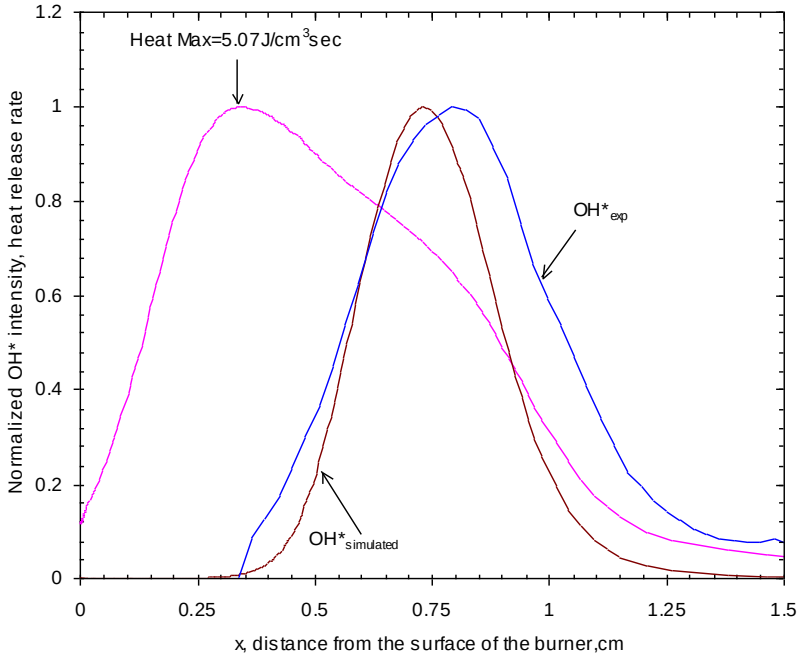
- 9%O_2/91%N2, m = 0.078 g/s, into 100%H_2 at 0.079 atm
- Xybion IMC-201 intensified multispectral video camera with filter
- OH* chemiluminescence



UV Imaging

- Comparison with computational modeling

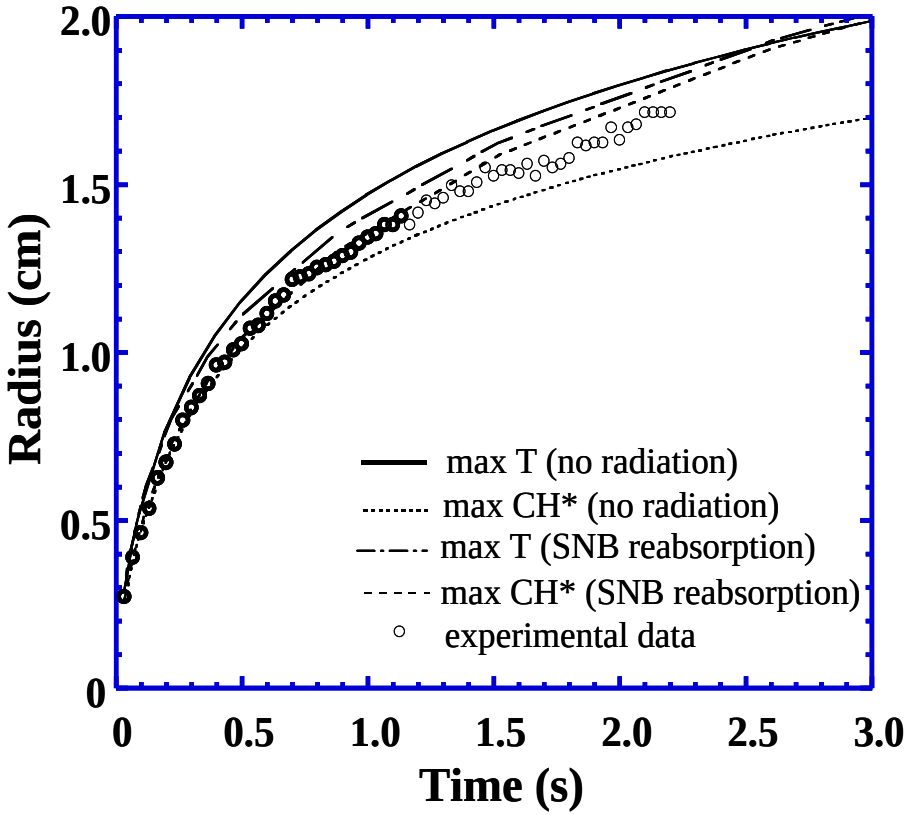






SDEP A2

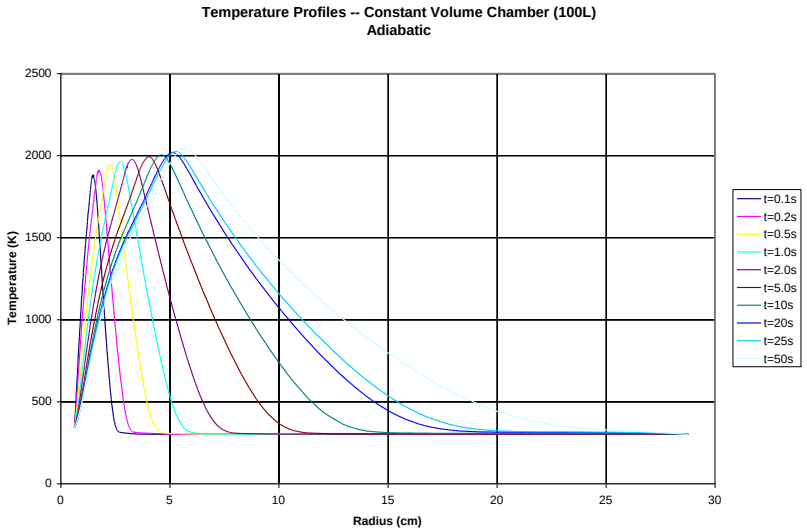
- Visible flame radius as a function of time, flow rate, and fuel composition.
- Requirements
  - Color images
  - UV images

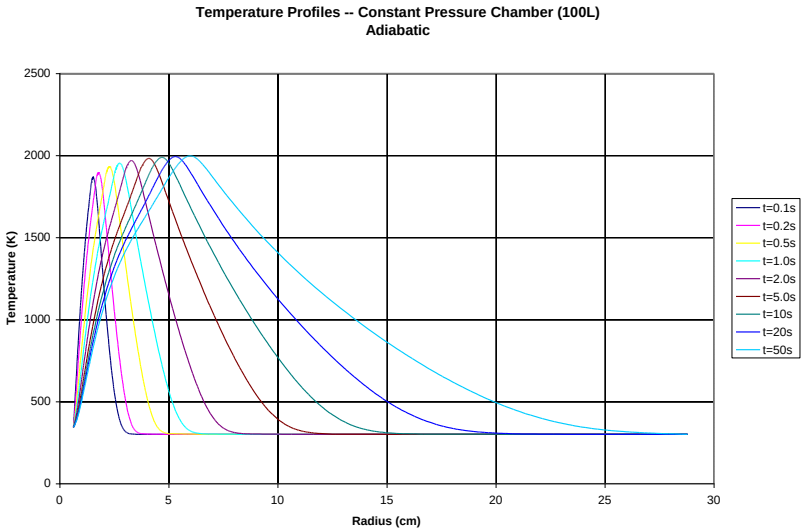




SDEP A3

- Temperature (and desired major species) radial distribution as a function of time, flow rate, and fuel composition.
- Requirements
  - IR images
  - Temperature profiles





20%H_2-25%CH_4-55%N_2 into air, m=15.72mg/s



SDEP A4

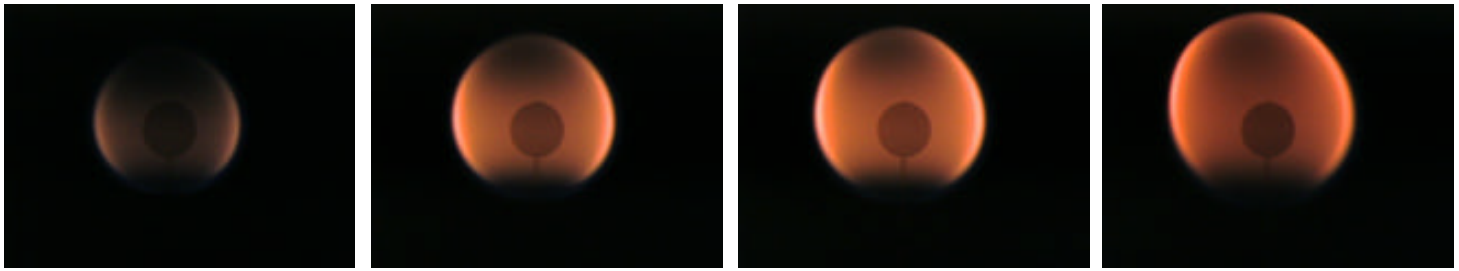
- Soot onset (determined from visible luminescence versus soot extinction measurements) as a function of radii for different flow rates and fuel compositions.
- Requirements
  - Color images
  - Soot volume fraction measurements
  - Temperature profiles



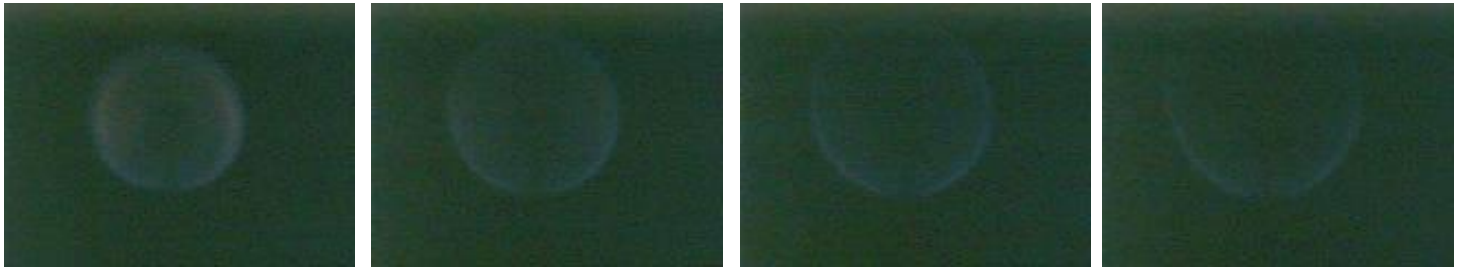
Sooty Formation

2.2-s Drop Tower Experimentation

- 25%C_3H_8 / 2%O_2 / 49% Ar / 24% N_2



High flow rate



Low flow rate



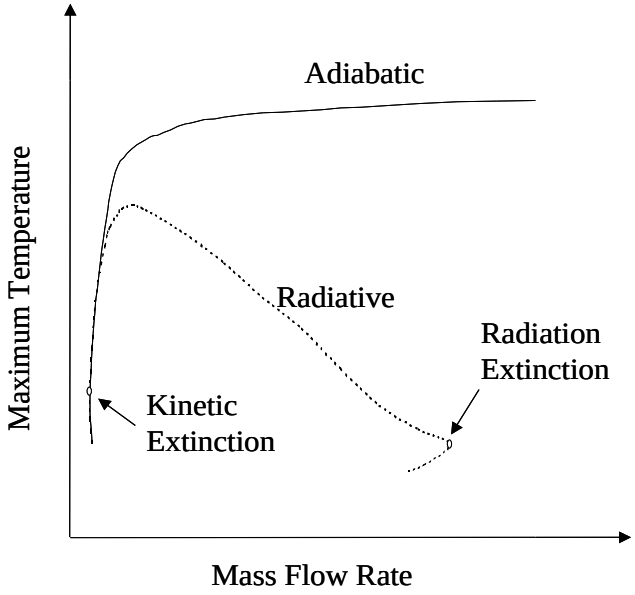
SDEP

Objective B) Dual Extinction Limits at Low and High System Da: Determine the quasi-steady convective/chemical-kinetic extinction limits and radiative/chemical-kinetic extinction limits for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty).



SDEP B1, B2, B4

- B1) Extinction limits as a function of system Damkohler number.
- B2) Total radiative loss as a function of radius for different flow rates and fuel compositions.
- B4) Radiation loss versus time for sooty flames, for different flow rates and fuel compositions.
- Requirements
  - Temperature profiles
  - Color images
  - UV images
  - IR images (desired)
  - Mass flow rate
  - Radiometer



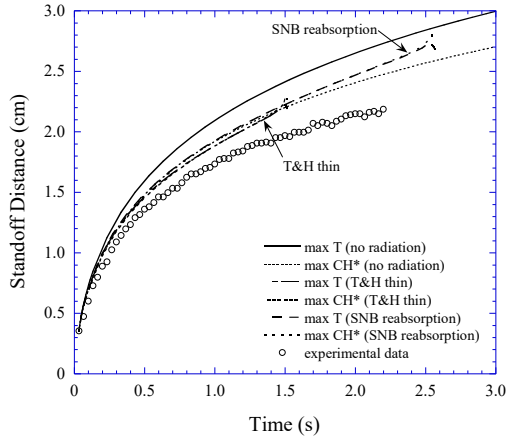


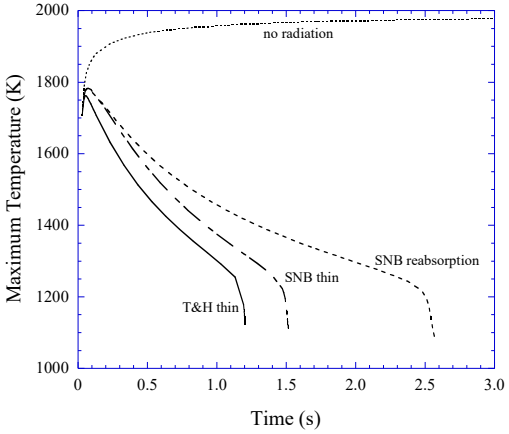
SDEP B3

- Peak CH* or OH* chemiluminescence along with peak temperature as a function of radius, for different times, flow rates, and fuel compositions.
- Requirements
  - UV images (CH* filter)
  - Temperature profiles

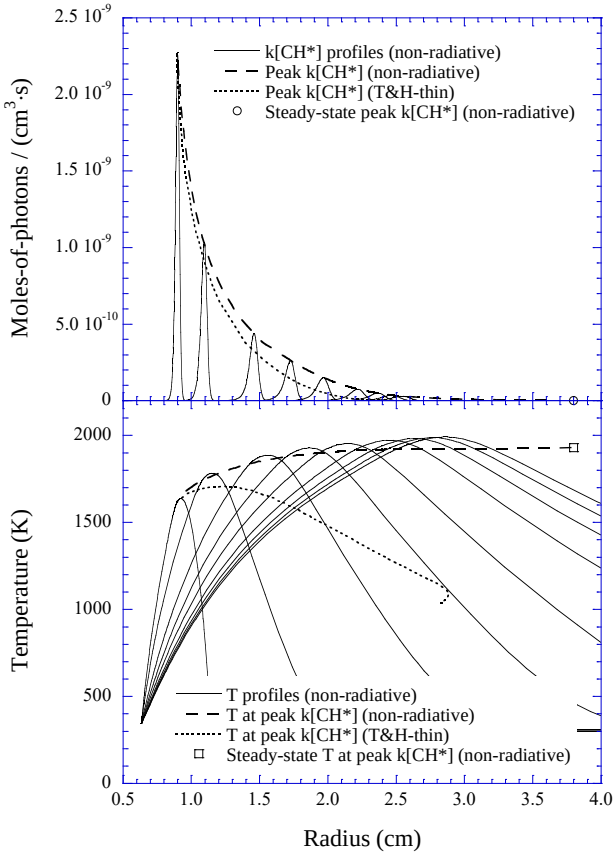


Extinction – Quasi-Steady





Visible Luminosity Consideration

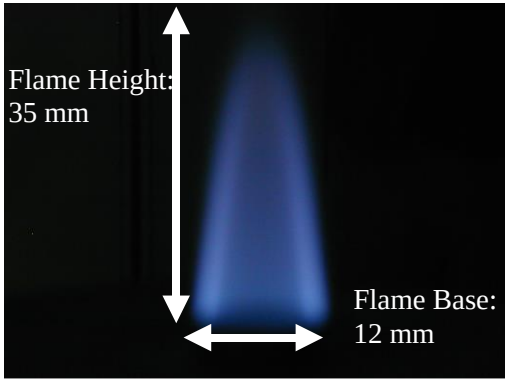


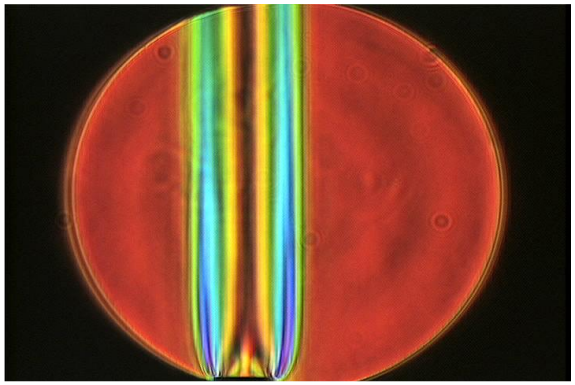
- $20\%H_2$-$25\%CH_4$-$55\%N_2$ (15.72mg/s) into STP air



1-D Gas-Phase Temperature (or Density) Measurement

- Assess evolving flame structure
- Assess moment of LOCAL flame extinction (based on precipitous drop in local peak temperature)
- Track flamefront motion based on peak temperature
- Rainbow schlieren preferred, but not required if radial temperature profile can be otherwise provided, e.g. thin filament pyrometry





20%H_2/25%CH_4/55%N_2 coflow in normal-*g*



SDEP

Objective C) Flame Front Instabilities in Non-Premixed Flames: Determine the existence, onset, and nature of pulsating instabilities theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Le.



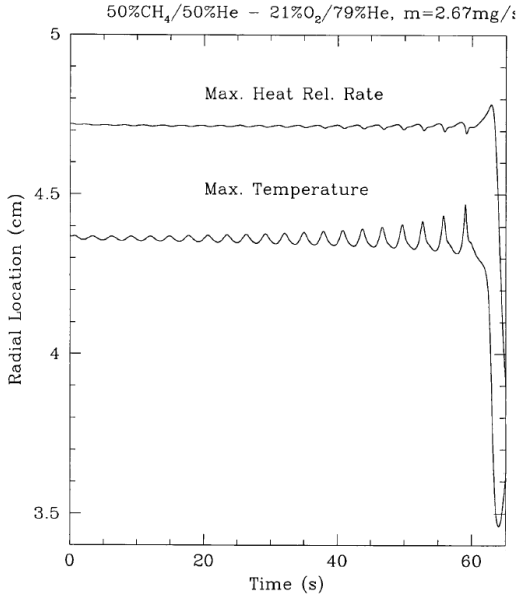
SDEP C1, C2, C3, C4

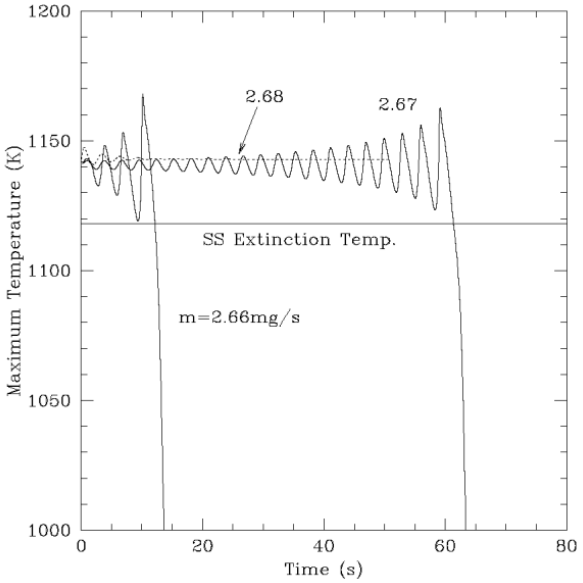
- C1) Total chemiluminescence as a function of time.
- C2) Total radiative loss (based on radiation flux) as a function of radius, for different flow rates and fuel compositions.
- C3) Oscillation mode, frequency, and amplitude near extinction as a function of flow rate for flames of different Le and mass flow rates.
- C4) Comparison of extinction times and extinction radii as a function of flow rate, for different fuel compositions
- Requirements
  - PMT (chemiluminescence)
  - Radiometer
  - Color images
  - UV images
  - IR images (desired)
  - Temperature profiles
  - Mass flow rate



Radiation Flux Measurement

- Global/local flame radiation measured at faster rate than cameras to capture “high” frequency oscillation





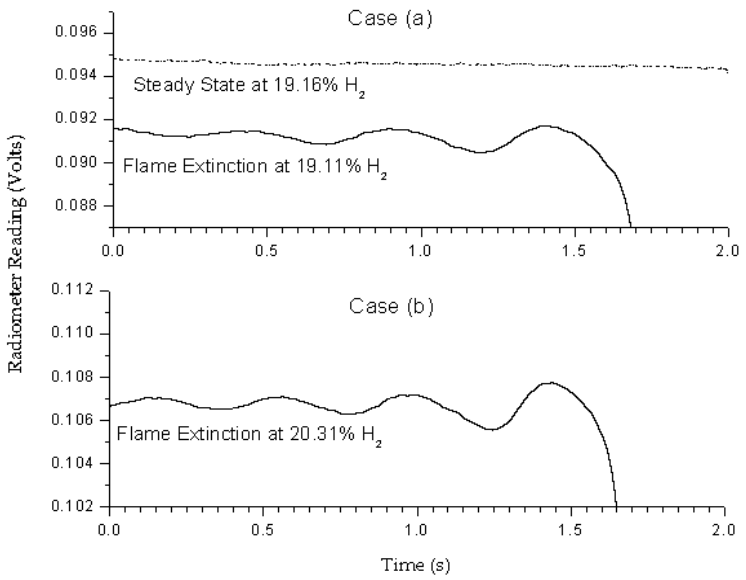
- Assess radiative heat transfer
- Flame movement often small
- Help to assess extinction moment



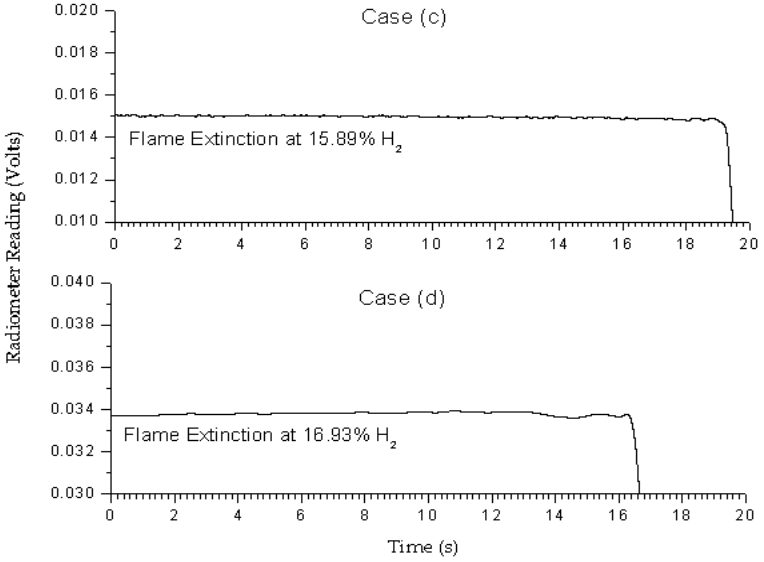
Extinction - Oscillatory

Micro-buoyancy Experimentation (Fuel: Hydrogen)

Radiation Extinction



Kinetic Extinction





Characteristic Flame Structure

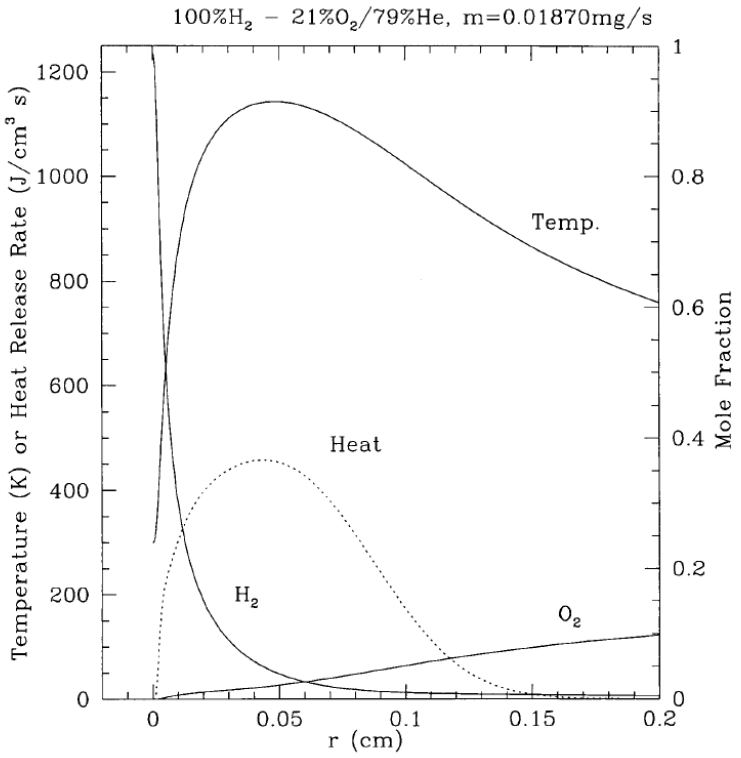


Fig. 15. Structure of a hydrogen diffusion flame with a mass flow rate of 0.018105 mg/s.

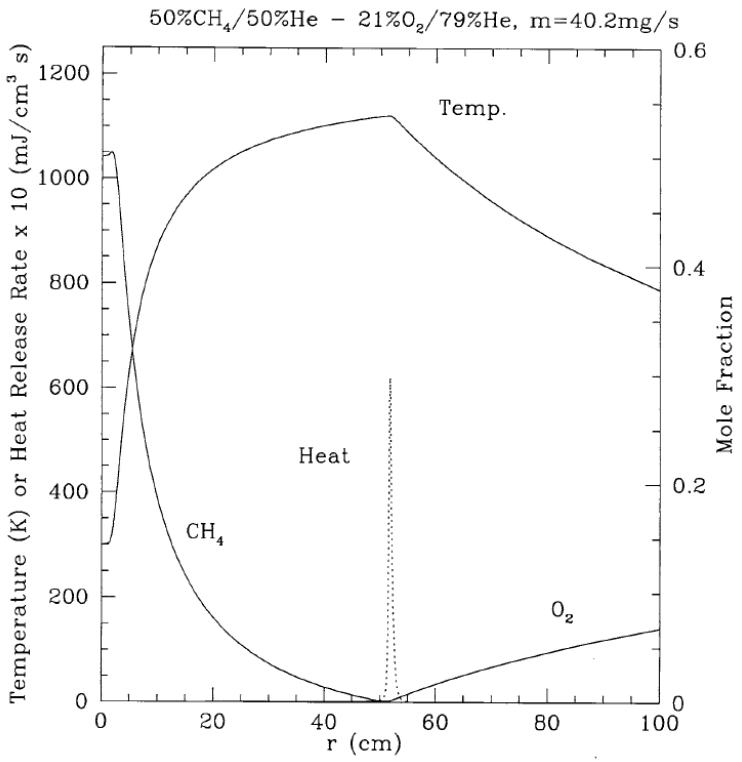


Fig. 16. Structure of a methane diffusion flame with a mass flow rate of 40.2 mg/s.

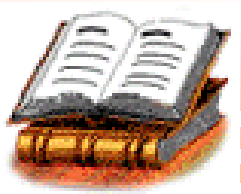


SDEP C5

- Soot radial distribution as a function of time for different flow rates and fuel compositions during fully transient flame front movement
- Requirements
  - Color images
  - Temperature profiles
  - Soot volume fraction measurements



Other Requirements





Burner (1/3)

- PI provided spherical porous bronze burner can presently provide 0.99 sphericity and 0.94 concentricity diffusion flames.



50%H_2/10%CH_4/40%N_2, 8.1mg/s into STP air (2.2-s drop tower), ½" burner diameter



Burner (2/3)

- Burner purpose is only to provide uniform, spherical flow

The classical thin-flame approximation is invoked such that, at the flame sheet, fuel and oxidizer are consumed completely and in stoichiometric proportion, as prescribed by the stoichiometric fuel/oxygen ratio f. Thus the flow field consists of an infinitesimally-thin flame, located at r_f, separating the chemically-frozen fuel and oxidizer regions. Diffusion of species into a spherical burner of porosity ϕ is allowed for, except at r=0 where the fuel mass fraction is fixed at Y_{FR} .

$$0 \le r \le R_b: \qquad Y_F = Y_{FR} + \left(\frac{f Y_{O\infty}}{1 - e^{-\frac{\Lambda_O}{r_f}}} \cdot \left(e^{-\frac{\Lambda_O}{r_f}} - e^{\frac{(\Lambda_F - \Lambda O)}{r_f}} \cdot e^{-\frac{\Lambda_F}{R_b}} \right) - Y_{FR} \right) \cdot \frac{e^{-\frac{\gamma}{r}}}{e^{-\frac{\gamma}{R_b}}}$$

$$R_b \le r \le r_f: \qquad Y_F = \frac{f Y_{O\infty}}{1 - e^{-\frac{\Lambda_O}{r_f}}} \cdot \left(e^{-\frac{\Lambda_O}{r_f}} - e^{\frac{(\Lambda_F - \Lambda O)}{r_f}} \cdot e^{-\frac{\Lambda_F}{r}} \right)$$

$$r_f \le r \le \infty: \qquad Y_O = \frac{Y_{O\infty}}{1 - e^{-\frac{\Lambda_O}{r_f}}} \cdot \left(e^{-\frac{\Lambda_O}{r}} - e^{-\frac{\Lambda O}{r_f}} \right)$$

$$\gamma = \dot{m}_s / 4\pi\phi\rho D_F$$

$$\Lambda_F = \dot{m}_s / 4\pi\rho D_F$$

$$\Lambda_O = \dot{m}_s / 4\pi\rho D_O$$

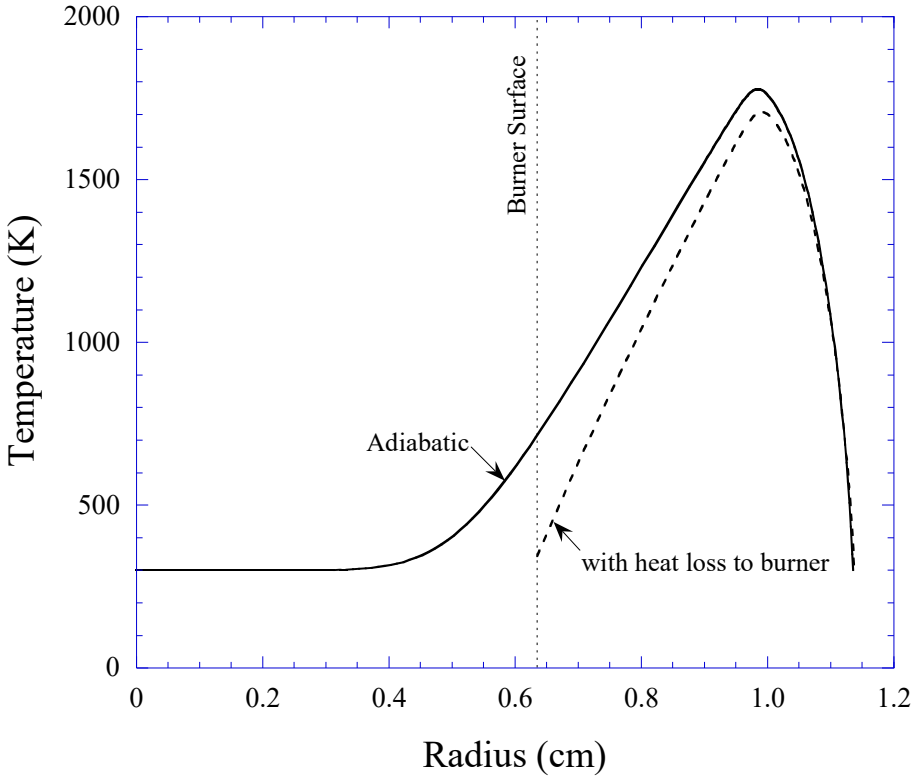
Independent of burner size and porosity ($r_f > R_b$)

$$r_f = \frac{\Lambda_O}{\ln\left(\frac{f Y_{O\infty}}{Y_{FR}} + 1 \right)}$$



Burner (3/3)

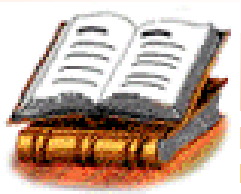
- ¼” (0.635cm) diameter burner preferred to minimize burner effect (i.e. achieve small flame dimensions, reduce heat loss to burner)



20%H_2-25%CH_4-55%N_2 into air, m=15.72mg/s

s-Flame

Operation Sequence





Experiment Plan (1/2)

I. Establish chamber at constant pressure and species concentration

II. Discharge fuel mixture through porous burner at initial flow rate

III. Ignite fuel mixture and retract upon successful ignition

IV. Allow for spherical diffusion flame to evolve

V. Apply one of the following:

- Maintain flow rate with same initial mixture for duration of experiment (25s) or until extinction
- Single step decrease in flow rate with same initial mixture at specified time, maintaining new flow rate for duration of experiment (25s) or until extinction

VI. Terminate fuel supply to burner after 25s.



Experiment Plan (2/2)

Prioritization and Logic

a) Transient phenomenon leading toward steady flame

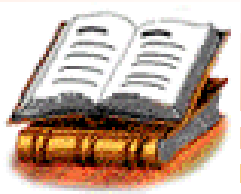
b) Extinction phenomenon at “quasi-steady”-states

c) Instability phenomenon, e.g. Le > 1 oscillations

→ Non-Sooty and Sooty flames



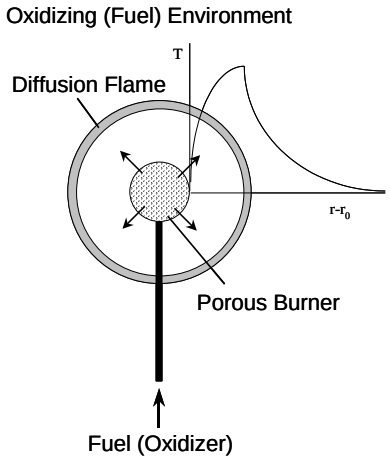
Test Matrix

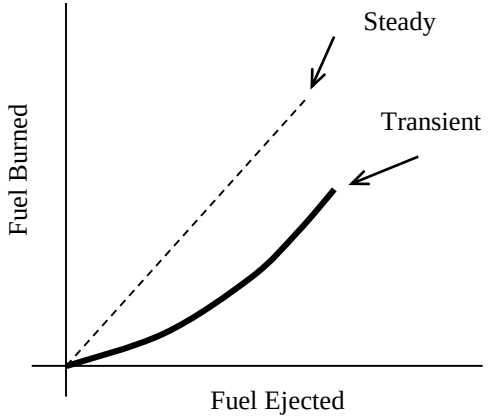




Experiment Setup

- All experimental sets will address transient phenomenon, as flame evolves from localized ignition towards “steady” behavior





Flame spread behavior

- Various mixtures to assess both chemical and transport effects on flame behavior



Test Matrix (1/4)

REQUIRED

Exp Set	Sub Set	Fuel Mixture	Ambient	Flow rate (cc/s)	No. of runs
1	(a)	25%H_2/20%CH_4/55%$\mathbf{N_2}$	21%O_2/79%$\mathbf{N_2}$	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s) (iii) 15 (entire duration, 25s) (iv) 10 (5s)→2 (rest of duration) (v) 5 (5s)→2 (rest of duration)	5
	(b)	20%H_2/25%CH_4/55%$\mathbf{N_2}$		(i) through (v) as shown above	5
	(c)	30%H_2/15%CH_4/55%$\mathbf{N_2}$		(i) through (v) as shown above	5
	(d)	45%H_2/55%$\mathbf{N_2}$		(x) 1 (entire duration, 25s) (y) 2 (entire duration, 25s) (z) 2 (5s)→0.5 (rest of duration)	3
	(e)	45%CH_4/55%$\mathbf{N_2}$		(i) through (v) as shown in (a)	5



Test Matrix (2/4)

REQUIRED Continued…

2	(a)	25%H_2/20%CH_4/55%**He**	21%O_2/79%**He**	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s) (iii) 15 (entire duration, 25s) (iv) 10 (5s)→2 (rest of duration) (v) 5 (5s)→2 (rest of duration)	5
	(b)	20%H_2/25%CH_4/55%**He**		(i) through (v) as shown above	5
	(c)	30%H_2/15%CH_4/55%**He**		(i) through (v) as shown above	5
	(d)	45%H_2/55%**He**		(x) 1 (entire duration, 25s) (y) 2 (entire duration, 25s) (z) 2 (5s)→0.5 (rest of duration)	3
	(e)	45%CH_4/55%**He**		(i) through (v) as shown in (a)	5
					Total: 46



Test Matrix (3/4)

HIGHLY DESIRED

Exp Set	Sub Set	Fuel Mixture	Ambient	Flow rate (cc/s)	No. of runs
3	(a)	20%C_2H_4/80%**N_2**	21%O_2/79%**N_2**	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s)	2
	(b)	25%C_2H_4/75%**N_2**		(i) and (ii) as shown above	2
	(c)	30%C_2H_4/70%**N_2**		(i) and (ii) as shown above	2
4	(a)	20%C_2H_4/80%**He**	21%O_2/79%**He**	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s)	2
	(b)	25%C_2H_4/75%**He**		(i) and (ii) as shown above	2
	(c)	30%C_2H_4/70%**He**		(i) and (ii) as shown above	2
					Total: 12



Test Matrix (4/4)

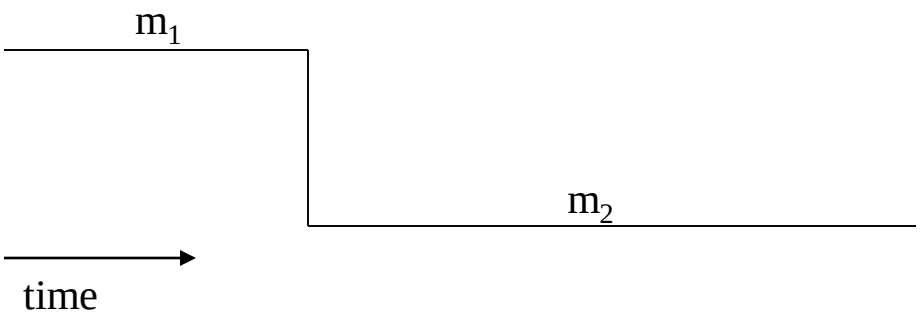
DESIRED

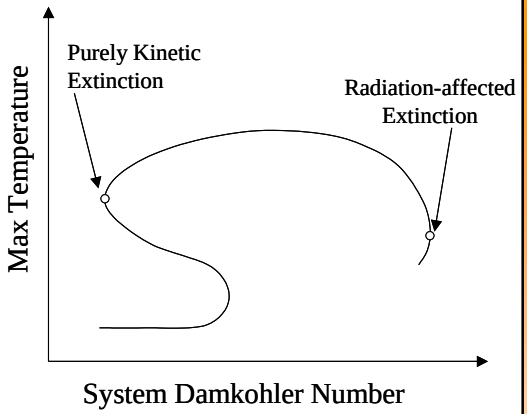
Exp Set	Sub Set	Fuel Mixture	Ambient	Flow rate (cc/s)	No. of runs
5	(a)	25%H_2/20%CH_4/55%$\mathbf{CO_2}$	21%O_2/79%$\mathbf{CO_2}$	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s) (iii) 15 (entire duration, 25s) (iv) 10 (5s)→2 (rest of duration) (v) 5 (5s)→2 (rest of duration)	5
	(b)	20%H_2/25%CH_4/55%$\mathbf{CO_2}$		(i) through (v) as shown above	5
	(c)	30%H_2/15%CH_4/55%$\mathbf{CO_2}$		(i) through (v) as shown above	5
	(d)	45%H_2/55%$\mathbf{CO_2}$		(x) 1 (entire duration, 25s) (y) 2 (entire duration, 25s) (z) 2 (5s)→0.5 (rest of duration)	3
	(e)	45%CH_4/55%$\mathbf{CO_2}$		(i) through (v) as shown in (a)	5
6	(a)	20%C_2H_4/80%$\mathbf{CO_2}$		(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s)	2
	(b)	25%C_2H_4/75%$\mathbf{CO_2}$		(i) and (ii) as shown above	2
	(c)	30%C_2H_4/70%$\mathbf{CO_2}$		(i) and (ii) as shown above	2
					Total: 29



Flow Supply and Flow Rate

- Deliver fuel, inert(s) to burner
- Different flow rates
  - Small versus large characteristic flame sizes
  - Ignite at moderate flow rate, establish quasi-steady flame, step flow rate down to examine kinetic extinction at small flame radii





- Different compositions as specified by Test Matrix
- Provide gas at max flow rate for at least 25s



Flow Supply and Flow Rate

- Different flow rates
  - Affect characteristic flame size (residence time)
    - Radiative extinction
    - Characteristic time to steady state
  - Kinetic extinction versus radiative extinction
- Step in mass flow rates are utilized
  - Ensure ignition and establishment of quasi-steady behavior
  - Inward flame evolution toward kinetic extinction



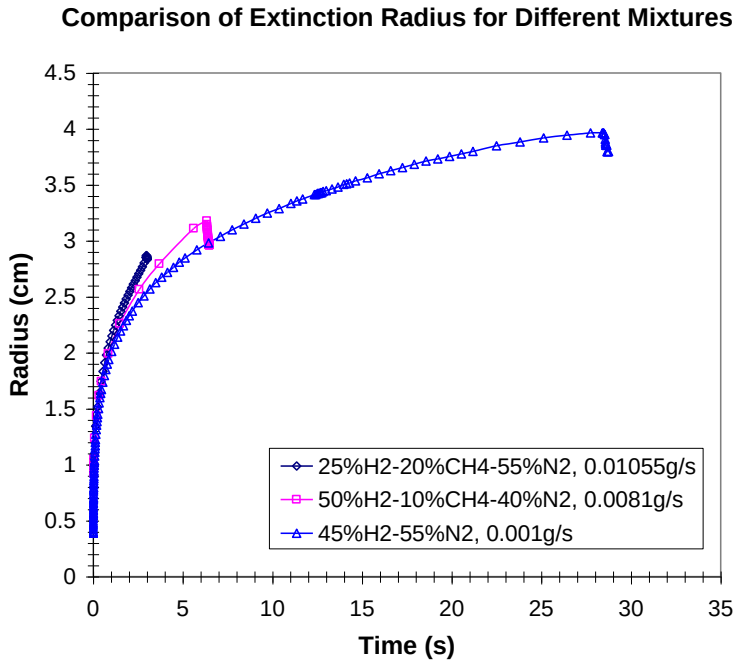
Fuel Mixtures

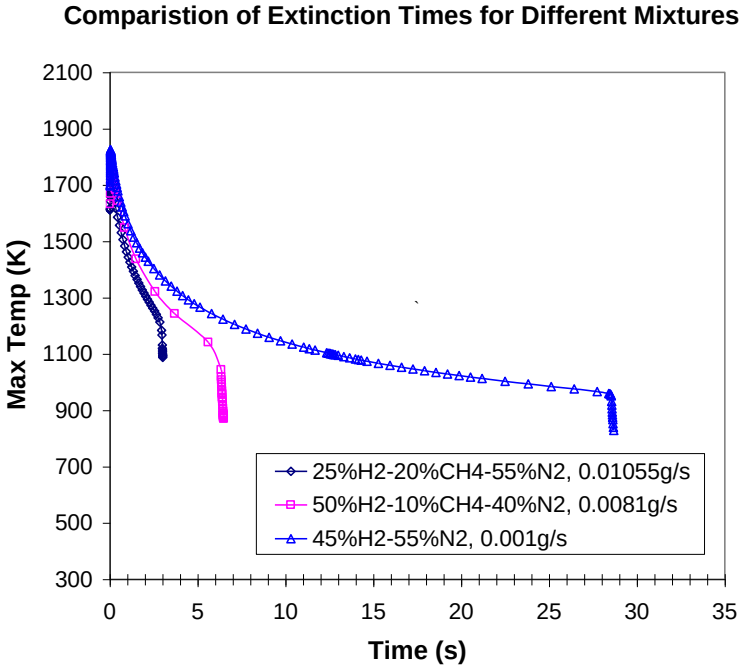
- H_2/Inert (**Required**)
  - Addresses pure hydrogen chemistry
  - Only H_2O in radiation considerations
  - Only OH* chemiluminescence
- CH_4/Inert (**Required**)
  - Addresses methane chemistry, e.g. GRI-Mech
  - H_2O, CO_2, CO in radiation considerations
- H_2/CH_4/Inert (**Required**)
  - Addresses chemical kinetics of blended mixture including kinetic extinction
  - Different characteristic radiative extinction times
  - OH* and CH* chemiluminescence



Fuel Mixtures

- 20%H_2/25%CH_4/Inert, 25%H_2/20%CH_4/Inert, 30%H_2/15%CH_4/Inert
  - Addresses chemical kinetics of blended mixture including kinetic extinction
  - Different characteristic radiative extinction times







Fuel Mixtures

- C_2H_4 (**Highly Desired**)
  - Nominally sooty C2 fuel
  - Sooting tendencies depend on characteristic residence time, so different flow rates must be examined



Test Matrix Rationale

- Different inerts to affect diffusive properties, radiative properties, and flame temperature
  - N_2 (**Required**)
    - Moderate flame temperature
    - Radiatively transparent
    - Normal “air”
  - He (**Required**)
    - Higher flame temperature
    - Radiatively transparent
    - High-Lewis number promoting pulsating instability
  - CO_2 (**Desired**)
    - Lower flame temperature
    - Optically participating medium (highly reabsorptive species)
- All inerts fixed at 0.55 mole fraction for non C_2H_2 fuel mixtures



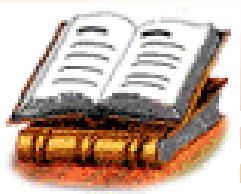
Ambient Environment

- STP
- Different sets of ambient gas mixtures to alter characteristic flame temperatures, diffusive transport, and radiative properties
  - "Air": 0.21 O_2, 0.79 N_2 (**Required)**
  - "He-Air": 0.21 O_2, 0.79 He (**Required**)
  - "CO_2-Air": 0.21 O_2, 0.79 CO_2 (**Desired**)
- Specified to within 1% mole fraction

- Ambient gas may be scrubbed and recycled for subsequent tests

s-Flame

Success Criteria





Minimal Success

- Observations of transient flame phenomena leading toward steady state flames for 1 CH_4 case and 1 C_2H_4 case; or
- Observations of flame extinction at both low and high system Damkohler number for 1 case; or
- Observations of spherical soot formation in 1 C_2H_4 case.



Significant Success

- Observations of at least two instances of flame front instabilities including at least one hydrocarbon fuel; or

- Combinations or extensions (in terms of number of mixture ratios for a single hydrocarbon fuel or number of hydrocarbon fuels) of at least two of items



Complete Success

- Observations of transient flame phenomena leading toward steady state flames for at least three different mixture ratios of each of two hydrocarbon fuels and two diluents; and

- Observations of flame extinction at both low and high system Damkohler number for at least three different mixture ratios of each of two hydrocarbon fuels and two diluents; and

- Observations of at least two instances of flame front instabilities each including at least one hydrocarbon fuel; and

- Observations of soot formation.